Omega: Continuous Creation, a Primordial Resonance, and the Local Emergence of General Relativity

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Abstract

Omega is a cosmological and gravitational framework in which the fundamental entity is not spacetime but a condensate that is continuously created. Spacetime, general relativity (GR) and relative time appear only locally, as properties of the perturbations of this medium. The universe has no age and no Big Bang: creation is eternal and proceeds in distinct episodes. No dark-matter particle and no dark-energy fluid are introduced.

The present formulation rests on four ingredients defined on a flat background with a universal time: (i) a tensor condensate $C_{\mu\nu}$ whose Goldstone modes are the gravitons and whose timelike eigen-direction is a universal clock; (ii) a dilation potential whose rolling produces creation at a rate Γ with zero total energy; (iii) a second field Ψ whose spatial gradient carries the low-acceleration (MOND-like) response and whose rhythm along the clock carries a pressureless component fixed by co-creation with baryons; (iv) a finite relaxation time of the response of Ψ.

We derive the local limit exactly (the Schwarzschild metric follows from the superposition of flow energies, giving the perihelion advance of Mercury and $\gamma = 1$); the exponent $3/2$ of the low-acceleration response imposed by flat rotation curves and the baryonic Tully–Fisher relation $V^4 = GMa_0$; the constraints imposed by Saturn and by the solar quadrupole measured by Cassini (a fast-saturating transition and a memory time $\tau \approx 0.15$–$0.5\,\mathrm{Myr}$); a de Sitter expansion at rate $H_\infty = \Gamma/3$ from a rolling vacuum at zero total energy; a periodic cycle of expansion, turnaround, compression and bounce; and the identity linking the pressureless component, the range of its mass term and the deviation of its rhythm from equilibrium.

The framework reproduces the galaxy data with $a_0 = 1.16 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, the high-redshift rotation curves (which exclude $a_0 \propto H(z)$ at $\Delta\chi^2 = +83$), the expansion history, the one-dimensional Lyman-α power measured by DESI, the black-body CMB and its acoustic structure. Its main open problems are declared: the coefficient κ in $a_0 = \kappa c\Gamma$ is not derived, the amount of the Ψ component is taken from data, galaxy clusters require a capture mechanism not yet found, and the baryonic Tully–Fisher slope is in tension with our own fit.

The paper ends with a dated table of falsifiable predictions. One of them is specific to Omega and free of adjustable parameters: if a_0 and the expansion share the same source, the ratio of a_0 in the distant past to its present value is fixed by the present equation of state of the vacuum, $a_0(z \gtrsim 1)/a_0(0) = \sqrt{1 + \frac{3}{4}(1 + w_0)}$.

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1 Introduction and epistemic status

1.1 Motivation

The standard cosmological model (ΛCDM) describes the expansion history, the cosmic microwave background (CMB) and large-scale structure with six parameters, at the price of two components that have never been observed directly: cold dark matter and dark energy. On galactic scales, the observed dynamics follow a tight relation between the observed acceleration and the acceleration produced by the visible baryons, with a single acceleration scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ [44, 42, 35]. That scale is numerically close to cH_0, a coincidence that ΛCDM does not explain and that modified-dynamics theories usually leave as a free constant.

Omega starts from the opposite end of the hierarchy. The fundamental object is a medium, the condensate, which is continuously created. General relativity is not the starting point but a local limit of the dynamics of this medium. The coincidence between a_0 and the expansion rate then becomes a structural requirement: both must come from the same source, the creation rate Γ.

1.2 Statement of the hypotheses

Omega rests on a small set of ontological statements that are not directly testable and that we label (G):

(H1) Before our dimension there exists a scalar field whose nature cannot be described with the concepts of our universe. A primary wave (a dimensional pulsation) crosses this field and leaves a resonance.

(H2) This resonance organises itself first at a sub-quantum level and unfolds into increasingly organised structures, up to the emergence of our universe. General relativity appears at this last level, and only locally.

(H3) The universe has no age. Creation is eternal and discontinuous: an unending sequence of episodes, with neither a first nor a last one. There is no Big Bang.

(H4) Time is a component of GR. It is therefore not a global coordinate of the underlying theory; the relative time of GR exists only where Omega reduces to GR.

(H5) Each class of cosmic body (star, galaxy, cluster, black hole) obeys its own rules; what decides is the set of forces concentrated at the boundary between the Omega regime and the GR regime.

1.3 Epistemic status: reverse engineering

The primordial field and the primary wave are accessible only through what they produced. Omega therefore reconstructs them by reverse engineering, starting from our universe. Any reconstruction fitted to existing data is partly circular. A framework of this kind becomes a theory in the strong sense only through predictions that are absent from the literature, written and dated before the measurement, and later confirmed. Section 17 gives these predictions; they were first recorded in a dated register on 21 and 22 October 2026.

1.4 Classification used throughout

Every statement carries one of the following labels.

- (O) raw observation;
- (M) data reduced with a model (often a GR-based model);

- (G) theoretical hypothesis of Omega;
- [D] result derived from the stated hypotheses and checked symbolically;
- [N] result established numerically (the scripts are available on request).

When a comparison uses a tool that presupposes GR at all scales (Boltzmann codes, standard growth equations), the tool is used to compare, never to conclude, except in regimes where Omega has been shown to reduce to GR (Sections 4 and 13).

1.5 What Omega does not introduce

- **No dark-matter particle.** No pressureless component is inserted by hand, and no ΛCDM parameter is used as an input of the theory. One pressureless component does appear (Section 9): it is the rhythm of a coherent field, not a population of particles, and it is required by a direct observation, the height of the third acoustic peak of the CMB. We state explicitly that, in the linear cosmological regime, it gravitates exactly like cold dark matter; it differs by its origin, by its fixed relation to the baryons, and by its behaviour in galaxies. Its amount is at present taken from data.
- **No dark-energy fluid.** The accelerated expansion is the dilation of the condensate at zero total energy (Section 8).
- **No inflation.** The primordial fluctuations come from the compression phase of the cycle (Section 12).

1.6 Plan

Section 2 sets out the foundations; Section 3 gives the present microscopic architecture. Sections 4 to 7 derive the local limit and the low-acceleration regime and confront them with the solar system, galaxies, groups and clusters. Sections 8 to 12 treat creation, the field Ψ, the cycle, the twin universe and the primordial fluctuations. Section 13 treats the CMB, Section 14 black holes and Section 15 the quantum level. Section 16 lists what is derived, what is postulated and what remains open. Section 17 closes with the table of predictions.

2 Foundations

2.1 The condensate and its clock (g)

The condensate is the fundamental medium. It evolves on a flat background, in a universal time: the time of its clock, the phase of the condensate, which advances everywhere at the same rate. This clock is the clock of permanent creation and defines a preferred frame. A possible, unproven coincidence is that this frame is the one in which the CMB dipole vanishes.

Since time is a component of GR (H4), it cannot be a global coordinate of a spacetime in Omega. A formal consequence follows: space and time do not have to play the same role in the action. Galactic behaviour can be fixed by the spatial variations of the condensate and cosmological behaviour by the rate of its clock. The closest mathematical structure in the literature is the *khronon* of Blanchet and Marsat [6], a field that defines a preferred time.

2.2 The emergent metric (g)

Small perturbations of a condensate propagate as if they saw a curved spacetime built from the density and the flow of the medium: this is the acoustic metric [60, 4]. In Omega, matter sees

this metric. In coordinates that follow the expansion of the condensate,

$$ds^2 = \Omega_c^2 \left[-c^2\, dt^2 + b^2\, d\vec{x}^{\,2}\right], \qquad \Omega_c^2 \propto \frac{\rho}{c_s}, \tag{1}$$

where b measures the expansion of the condensate. Redshift and distances follow from this metric alone.

This programme belongs to the family of induced gravity (Sakharov [56]), of Volovik's analogue of the universe in superfluid helium [63], and of the synthesis of Barceló, Liberati and Visser [4]. Their robust result is that geometry emerges easily; their known difficulty is to obtain the Einstein dynamics itself. The universal time of Omega breaks Lorentz invariance at the fundamental level, which bypasses the main theoretical obstacle, the Weinberg–Witten theorem [66]. Section 3 shows how the full GR dynamics is recovered from a tensor condensate.

2.3 The hierarchy: GR as a local limit

- At high acceleration ($g \gg a_0$), Omega reduces to Newton and to GR.
- At low acceleration, Omega governs.
- The solar system, the Moon and binary pulsars are therefore not "GR imposing its rules on Omega". They are checks that Omega gives back GR where it must.
- The boundary between the two regimes is a physical surface: around any body, it is the surface on which the field gradient that carries the low-acceleration response passes from its screened (GR) regime to its unscreened regime, that is, where the acceleration is of order a_0.

2.4 Cosmic bodies and the Omega/GR boundary (g)

There is no single universal ingredient. Each class of body (star, galaxy, group, cluster, black hole) follows its own rules, and the forces concentrated on the Omega/GR boundary decide which kind of structure develops. This statement is not a way of avoiding prediction: it has a precise counterpart in the field Ψ (Section 9), whose response depends on its local density and therefore differs from one class of body to another.

2.5 The twin universe (g)

Our universe and a twin universe form a polarised pair, like the two poles of a battery. The twin drives creation in our sector. Section 11 gives the constraints that the observed constancy of the creation rate imposes on this pair.

2.6 The CMB as the echo of creation (g)

The CMB is the echo of the continuous creation of our dimension, which precedes the appearance of GR. It is not a frozen image: it is the resonance of perpetual creation, still in motion. Section 13 shows what the measurements say about its origin and what targets its structure sets for the theory.

3 The microscopic architecture

This section gives the present architecture of Omega as a whole. The following sections derive its consequences one by one.

3.1 Overview

On a flat background with a universal time t:

1. **A tensor condensate** $C_{\mu\nu}$ (cardinal gravity). Its Goldstone modes are the gravitons and give back GR at low energy. Its timelike eigen-direction u^μ is the universal clock. Its spatial eigenvalues $A(t)$ carry creation, with a dilation potential

$$V(A) = -\tfrac{3}{2}\, m^2 A^2 + \tfrac{3}{2}\,\lambda A^4 + \tfrac{3}{2}\,\frac{\varepsilon}{A^2}, \qquad m = \tfrac{2}{3}\,\Gamma. \tag{2}$$

   The three terms produce, respectively, creation (Section 8), the turnaround of the cycle and the bounce (Section 10).

2. **A field** Ψ described by two variables,

$$Q \equiv u^\mu \partial_\mu \Psi, \qquad Y \equiv \left|\nabla_\perp \Psi\right|^2, \tag{3}$$

   its rhythm along the clock and its gradient perpendicular to the clock. Its energy density is a function $\mathcal{F}(Y, Q)$ with three regimes: near an equilibrium rhythm Q_0 in cosmology, a stiffening function of $Q - Q_0$, which produces a pressureless component; at large gradient, a steep term that screens Ψ (Newton and GR); at small gradient, a term $\propto Y^{3/2}/a_0$, which produces the low-acceleration response. A small mass term μ completes the static limit. The response of Ψ relaxes towards its equilibrium value with a finite time τ.

3. **Matter** couples universally to the emergent metric.

3.2 The tensor condensate: cardinal gravity

Kostelecký and Potting [30] start from a symmetric tensor field $C_{\mu\nu}$ on a flat Minkowski background, without dynamics of its own. A potential gives C a non-zero vacuum value that breaks Lorentz invariance. The gravitons are the Goldstone modes of this broken order: six modes, which reproduce linearised gravity and its two tensor polarisations exactly. By coherent coupling to energy (bootstrap), they recover the Einstein–Hilbert action well below the Planck scale. Four additional massive modes remain. Gravitons as Goldstone modes were also studied by Bjorken and by Kraus and Tomboulis [32].

This is structurally the hierarchy of Omega: a condensate on a flat background, GR as its low-energy limit. The vacuum value has a timelike eigen-direction, which defines a preferred time; in Omega, it is the clock of the condensate.

What the tensor provides	Role in Omega
Its ripples (Goldstone modes)	gravitons, two polarisations, GR at low energy
Its timelike eigen-direction u	the universal time
Its spatial eigenvalues A	the spatial metric, $a^2 \propto A$; creation is their growth
Its potential $V(A)$	creation, turnaround and bounce

Table 1: Roles of the tensor condensate.

Quantitative requirements, all satisfied by construction or by bounds on free parameters:

1. The four additional modes must be massive: above 10^{-12} eV so as not to be radiated by the double pulsar or seen by LIGO; above about 5 meV if they couple with gravitational strength, because torsion balances exclude a Yukawa force of gravitational strength beyond a few tens of micrometres [33].

2. Any scalar ("sound") emission channel of the clock must stay below 10^{-4} of the quadrupolar emission (double pulsar, [31]).
3. Preferred-frame residuals: $\alpha_1 < 10^{-5}$, $\alpha_2 < 4 \times 10^{-7}$ (Section 4.4).

3.3 Why a tensor and not a scalar condensate [d]

A condensate described only by a phase, a flow and quantised vortices produces an acoustic metric whose spatial part is flat: all of gravity is carried by g_{0i} (the flow) and g_{00} (the flow energy). A gravitational wave with polarisations $+$ and $\times$ is a transverse deformation of space. For a wave along z, the components h_{xx}, h_{xy}, h_{yy} are gauge invariant: no change of coordinates brings them back to a flat spatial part. A scalar condensate therefore has scalar and vector waves but no tensor wave. Observations exclude this:

Observation	Value
Polarisation of GW170817 (pure polarisations)	$\log_{10} B = 20.81$ against pure vector, 23.09 against pure scalar [40]
Double pulsar [31]	quadrupolar emission verified to 1.3×10^{-4} (95 %)
Speed of gravitational waves [39]	between -3×10^{-15} and $+7 \times 10^{-16}$ of c

Table 2: Observations that require tensor order in the condensate.

The scalar condensate is therefore excluded as the only source of the emergent metric; the order must be tensorial.

3.4 The field Ψ

One clock cannot carry both a pressureless component and the low-acceleration response (theorem of Section 9.2). A second field Ψ is required. Normalising Ψ as a gravitational potential (so that, in a static galaxy, its gradient is an acceleration), its Lagrangian density is

$$\mathcal{L}_\Psi = -\frac{1}{8\pi G}\,\mathcal{F}(Y,Q), \qquad \mathcal{F}(Y,Q) \simeq \begin{cases} K(Q) & \text{cosmology, } Y \to 0, \\ \dfrac{2}{3}\,\dfrac{Y^{3/2}}{a_0} & \text{small gradient (deep regime),} \\ \lambda_s\, Y & \text{large gradient (Newton/GR),} \end{cases} \tag{4}$$

with the precise forms given in Sections 5 and 9. In the static limit, the shift of the rhythm of Ψ inside a potential well produces a mass term μ (Section 9.5). This is exactly the structure of the scalar sector of the Aether–Scalar–Tensor theory (AeST) of Skordis and Złośnik [58], reached here by a different path: in AeST the timelike vector is a fundamental field; in Omega it is the eigen-direction of the tensor condensate, which also generates gravity and creation. What is specific to Omega is listed in Section 16.

3.5 Parameter count

4 Local emergence of general relativity

For a static, spherically symmetric field, Omega gives back GR exactly from the flow of the condensate, without assuming it. Gravitational waves additionally require that the order of the condensate be tensorial (Section 3).

Observed quantity	Value	From	Status
G	—	scale of the gravitational sector	input
$H_\infty = \Gamma/3$	$\Gamma = 5.46 \times 10^{-18}\,\mathrm{s}^{-1}$	m in $V(A)$	input (M)
a_0	$1.16 \times 10^{-10}\,\mathrm{m\,s}^{-2}$	scale of $\mathcal{F}$	relation $a_0 = \kappa c\Gamma$ postulated, κ not derived
Turnaround	$a_{\max} \geq 1.81$	λ	bounded
Preferred-frame residual	$\alpha \lesssim 10^{-7}$	clock coupling	bounded
Ratio ω_Ψ/ω_b	5.364 ± 0.065	Q_0 and the charge of Ψ	relation from co-creation; $\hbar Q_0$ not derived
Memory τ	0.15–0.5 Myr	relaxation of Ψ	input
Mass term μ	$\mu^{-1} \gtrsim 1$ Mpc today	$K''(Q_0)$	bounded
Additional modes	$\gtrsim 5$ meV	tensor potential	bounded

Table 3: Parameters of Omega at low energy. Without additional relations, Omega describes these numbers without predicting them; its parameter-free predictions are listed in Section 17.

4.1 Gravity as a flow of the condensate

In the acoustic metric, the gravity of a mass M is an inflow of the condensate towards it, with speed $\sqrt{2GM/r}$. Permanent creation adds an outward flow $H_\infty r$ everywhere.

What superposes [d]. If the two velocities simply added, a cross term would produce an extra acceleration towards the mass,

$$a_\times = H_\infty \sqrt{\frac{GM}{2r}}, \tag{5}$$

and a universal retrograde perihelion precession of $-0.53\,H_\infty$ per unit time, about -0.63 mas per century for every planet (the smallness of the effect is exactly compensated by the length of the orbit). Against the INPOP15a ephemerides (M) [28], this is 0.6σ for Mercury (0.0 ± 1.05) but 3.4σ for Saturn (0.05 ± 0.20 mas per century). The data therefore require that what superposes is not the flow velocities but the flow energies per unit mass, $v^2/2 = -\Phi$: Bernoulli-type fluid mechanics with Newtonian gravity. The condensate dynamics must be non-linear and not that of an ordinary fluid.

4.2 Derivation of the local limit [d] [n]

Three ingredients:

1. (G) matter sees the acoustic metric of the condensate;
2. (G) the density and sound speed of the condensate are constant, which permanent creation ensures;
3. (O)/(M) flow energies superpose: $v^2/2 = -\Phi$.

With a radial flow of velocity $v(r)$ (negative for an inflow), the acoustic metric reads

$$ds^2 = -c^2dt^2 + (dr - v\,dt)^2 + r^2d\Omega^2 = -(c^2 - v^2)\,dt^2 - 2v\,dr\,dt + dr^2 + r^2d\Omega^2. \tag{6}$$

Redefine the clock by

$$dt = dT - \frac{v\,dr}{c^2 - v^2}. \tag{7}$$

The cross terms cancel: $-2(c^2-v^2)\left(-\frac{v}{c^2-v^2}\right)dT\,dr - 2v\,dT\,dr = 0$. The dr^2 terms give

$$-\frac{v^2}{c^2-v^2} + \frac{2v^2}{c^2-v^2} + 1 = \frac{c^2}{c^2-v^2}. \tag{8}$$

The metric becomes static and diagonal:

$$g_{TT} = -(c^2-v^2) = -c^2\left(1+\frac{2\Phi}{c^2}\right), \qquad g_{rr} = \frac{c^2}{c^2-v^2}, \qquad g_{TT}\,g_{rr} = -c^2. \tag{9}$$

The identity $g_{TT}g_{rr} = -c^2$ holds for any potential Φ. For a point mass, $\Phi = -GM/r$, this is exactly the Schwarzschild metric: an algebraic identity, not an assumption. Direct numerical integration of orbits and light rays gives:

Test	Omega (derived)	Measurement
Perihelion advance of Mercury	$42.98''$ per century	$42.98 \pm 0.04''$ [68]
Light deflection at the solar limb	$1.751''$	$\gamma = 1 + (2.1 \pm 2.3) \times 10^{-5}$ (Cassini [5])
PPN parameter γ	exactly 1, for any Φ	idem

Table 4: Classical tests of the local limit. These are retrodictions shared with GR.

Writing Schwarzschild as "flowing space" is known (the river model of Hamilton and Lisle [25], the acoustic metric of Unruh [60]). What is specific to Omega is the reason: the superposition of flow energies imposed by the data, and the balance between creation and consumption of the medium.

Consequence: lensing follows dynamics at all accelerations. Since $g_{TT}g_{rr} = -c^2$ holds for any Φ, including in the low-acceleration regime, the mass seen by lensing equals the mass seen by velocities. TeVeS needed additional fields to achieve this; here it is automatic. This is consistent at the level of the broad trends with KiDS weak lensing [10].

The price. For the density to stay constant with an inflow $\sqrt{2GM/r}$, the condensate must be absorbed around masses at the rate $(3/2)\sqrt{2GM/r^3}$. For the Sun, this rate equals the creation rate at about 90 pc; beyond, creation dominates. In the tensor version, Schwarzschild follows from the field equations of the Goldstone sector and this consumption may no longer be needed.

4.3 Frame dragging

The gravitomagnetic effect of a rotating mass is the rotating part of the flow, $v \propto (\vec{J}\times\vec{r})/r^3$, whose vorticity is a dipolar field in $1/r^3$. A smooth scalar superfluid, whose flow derives from a single phase, has no vorticity and therefore no frame dragging, while Gravity Probe B measures 37.2 ± 7.2 mas/yr [20]. With quantised vortices the effect is recovered (predicted 39.2 mas/yr with the exact geometry of the orbit). In the tensor version, frame dragging comes from the Goldstone modes, as in linearised gravity, and no vortex density is needed. Its coefficient carries no new constant provided the dynamics is Lorentz invariant at the emergent scale: a moving mass is then the boosted version of a mass at rest. The same condition fixes α_1, α_2 and frame dragging.

4.4 Preferred-frame residuals

The universal time could leak into local physics. Order-of-magnitude residuals, with the last free coefficient of order one and the softest transition between the two regimes:

Place	Residual α_1	Bound	Residual α_2	Bound
Earth–Moon (lunar laser ranging)	4×10^{-7}	10^{-5}	4×10^{-8}	4×10^{-7}
Saturn (Cassini)	1.5×10^{-5}	10^{-5}	2×10^{-6}	4×10^{-7}
Solar surface, pulsars	$\lesssim 10^{-11}$	10^{-5}	$\lesssim 10^{-12}$	4×10^{-7}

Table 5: Estimated preferred-frame residuals.

With a faster transition, such as the one fitted to galaxies (Section 5), all residuals fall below 10^{-11}. The only real constraint is an upper bound on the residual of preferred time at high acceleration, $\alpha \lesssim 10^{-7}$. Blanchet and Marsat [6] point to a preferred-frame term already at Newtonian order, small if the universal time coincides with the CMB frame; it remains to be quantified against the ephemerides.

4.5 Universality of the emergent metric

All particles must see the same metric.

Requirement	Measurement (o)	Consequence
E1. Same limiting speed for all	GW170817: 10^{-15}	gravitational waves share the light cone exactly
E2. Invisible granularity	Crab nebula: no deviation up to $\sim 2 \times 10^{16}$ GeV [38]	healing length $\lesssim 10^{-32}$ m (about 600 Planck lengths): a near-Planckian condensate (assuming CPT and P)
E3. No leak of universal time into matter	precision tests on matter	suppression in $(M_*/M_{\mathrm{Pl}})^2$ [53]

Table 6: Universality requirements.

Universality (E1) cannot be proven without a microscopic model: Omega must posit it as a principle, the equivalent of the equivalence principle. Volovik [63] offers a lead, Fermi points, topological properties that would force all excitations to see the same metric. The condensate thus has two extreme scales: a microscopic one close to the Planck scale and a macroscopic one fixed by Γ.

4.6 Gravitational constant

In this framework G is fixed by the microscopic constituents of the condensate. The rolling vacuum (Section 8) changes the scale of space, not G. Omega therefore predicts $\dot{G} = 0$, consistent with lunar laser ranging, $\dot{G}/G = (7 \pm 8) \times 10^{-14}$ per year [26].

5 The low-acceleration regime

Below a_0, the response of the condensate changes exponent. The data fix this exponent, the shape of the transition and a memory time, independently of GR.

5.1 The criterion is an acceleration

In a weak static field, the gradient of Ψ normalised as a potential is an acceleration, and in the khronon formulation the acceleration of the timelike direction u is g/c^2 [6]. The criterion

"g compared with a_0" is therefore not a choice but a geometrical property of the fields of the condensate. Two consequences follow:

- there is no slip between the two gravitational potentials: lensing follows the low-acceleration potential (Section 4.2);
- on the homogeneous cosmic background, the spatial gradient of Ψ vanishes: the low-acceleration term acts only on fluctuations, in the quasi-static non-linear regime, never on the mean expansion.

For the primordial fluctuations, the mean acceleration is $13.8\,a_0$ at recombination, $1.1\,a_0$ at $z = 100$ and $0.23\,a_0$ at $z = 20$. Since the low-acceleration term is carried by the gradient of Ψ and acts only in the quasi-static non-linear regime, it does not apply to linear cosmological perturbations; at $x = g/a_0 = 14$ the deviation from Newton would in any case be only 0.5 % to 2.4 % depending on the transition.

5.2 The exponent imposed by flat rotation curves [d]

Assume that the condensate responds to the potential with a power of the gradient:

$$\nabla \cdot \left(\frac{|\nabla\Phi|^n}{a_0^n} \nabla\Phi \right) = 4\pi G\rho. \tag{10}$$

For a point mass M, Gauss's theorem gives $|g|^{n+1} r^2/a_0^n = GM$, hence

$$g \propto r^{-2/(n+1)}. \tag{11}$$

Flat rotation curves (O) require $g = V^2/r \propto 1/r$, hence $n = 1$ necessarily. The energy of the response grows as $|\nabla\Phi|^3$, that is, as the power $3/2$ of its kinetic invariant Y: this is the origin of the term $Y^{3/2}/a_0$ in Eq. (4). No other power gives flat curves.

With $n = 1$, $g^2/a_0 = GM/r^2$, so $g = \sqrt{GMa_0}/r$ and $V^2 = gr = \sqrt{GMa_0}$:

$$V^4 = G\,M\,a_0, \tag{12}$$

the baryonic Tully–Fisher relation with slope exactly 4. Our own fit of SPARC gives a slope of 3.62 ± 0.09, in tension with 4; Lelli et al. [36] find about 3.85 with the flat velocity. This tension is shared with every theory of the MOND family and is declared open.

5.3 Shape of the transition: two stages in series (g)

The gravitational field crosses two stages of the condensate, like a current through two resistors in series: a linear stage with constant conductance, which gives Newton; and a stage limited by creation, whose conductance grows with the field. The intuition is that below a_0 the renewal of the medium is faster than the time the field needs to establish itself over the relevant distance. With $x = g/a_0$ and the interpolating function $\mu_{\rm M}$ defined by $g_{\rm N} = \mu_{\rm M}(x)\,g$:

$$\frac{1}{\mu_{\rm M}} = 1 + \frac{1}{G_2(x)}, \qquad G_2(x) = x\,(1 + x^p). \tag{13}$$

The family was tested against SPARC and against the bound of about $9 \times 10^{-15}\,\mathrm{m\,s^{-2}}$ on an anomalous acceleration at Saturn (M):

The second stage must switch off quickly at high field: $p \geq 1$. With $p = 1$, SPARC stays well fitted (0.137 against 0.133 dex). The value of a_0, and therefore $\kappa = a_0/(c\Gamma)$, depends on the shape: $\kappa = 0.071$ with the McGaugh function, 0.095 with $p = 1$. These comparisons use Newton for the baryonic mass models (no GR); Saturn uses GR as the local reference, which is legitimate since the local limit of Omega has been derived.

p	fitted a_0 ($10^{-10}\,\mathrm{m\,s^{-2}}$)	SPARC scatter	excess at Saturn	verdict
0 ("simple" function)	2.29 (1.14 in the deep regime)	0.133 dex	1.1×10^{-10}	excluded ($\times 10^4$)
0.5	1.74	0.135 dex	2.8×10^{-13}	excluded
0.72	1.68	0.136 dex	1.6×10^{-14}	marginally excluded
1	1.56	0.137 dex	3.7×10^{-16}	allowed
2	1.43	0.140 dex	7×10^{-22}	allowed
McGaugh reference	1.16	0.133 dex	—	—

Table 7: Transition functions against SPARC and Saturn. Excesses in $\mathrm{m\,s^{-2}}$.

5.4 The solar quadrupole: memory is required [n]

The field of the Galaxy at the Sun ($1.37 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ for the Newtonian field of the baryons, about 2.2×10^{-10} observed (M)) distorts the low-acceleration regime around the Sun (external field effect) and creates a quadrupole Q_2 in the solar potential. Cassini radio tracking bounds it to $Q_2 = (1.6 \pm 1.8) \times 10^{-27}\,\mathrm{s^{-2}}$ [50].

The quadrupole is produced around the MOND radius of the Sun (about 7 000 AU), where the acceleration is of order a_0; it barely depends on the high-gradient shape of the transition:

Response (series, exponent p)	fitted a_0 ($10^{-10}\,\mathrm{m\,s^{-2}}$)	SPARC scatter	$\lvert Q_2\rvert$ without memory
McGaugh	1.16	0.133 dex	$3.2 \times 10^{-26}\,\mathrm{s^{-2}}$
$p = 1$	1.56	0.137 dex	2.7×10^{-26}
$p = 2$	1.43	0.140 dex	2.66×10^{-26}
$p = 4$	1.37	0.142 dex	2.59×10^{-26}
$p = 8$	1.37	0.144 dex	2.55×10^{-26}

Table 8: Solar quadrupole without memory, for increasingly steep transitions.

All exceed the measured value by a factor 17 to 20 and the 2σ upper bound by a factor 5 to 6. A stronger screening at large gradient changes nothing. **No instantaneous response passes Cassini.**

5.5 The memory time τ (g)

The response of Ψ has a finite relaxation time τ. The low-acceleration effect adjusts to its equilibrium value with a delay, through a causal relaxation equation of Maxwell–Cattaneo type,

$$\tau \frac{d\mathbf{g}_\Psi}{dt} + \mathbf{g}_\Psi = \mathbf{g}_\Psi^{\mathrm{eq}}, \tag{14}$$

where $\mathbf{g}_\Psi$ is the low-acceleration contribution along the trajectory. The effect is smeared along the motion of the Sun, over a wake of length $L = v\tau$, which suppresses Q_2.

- Wake constraint: $\tau \gtrsim 0.15\,\mathrm{Myr}$. Allowed window: $\boldsymbol{\tau \approx 0.15}$**–0.5 Myr**.
- The window passes Cassini, wide binaries, SPARC, the high-redshift rotation curves and the classical dwarf spheroidals.
- The memory cannot be a slowly propagating wave ($c_S \approx$ 65–220 m/s would produce gravitational Cherenkov radiation [19]): it is a local dissipative relaxation.

The condensate thus carries two characteristic times, both required by the data:

Time	Value	Fixes	Data
$1/\Gamma$, renewal	about 5.8 Gyr	a_0 and the expansion	expansion, galaxies
τ, memory	0.15–0.5 Myr	invisibility of the regime in the solar system	Cassini

Their ratio $\tau\Gamma \approx 3\text{–}9 \times 10^{-5}$ would become a prediction if τ were derived from the creation rate (Section 16).

5.6 Status

The series mechanism is a phenomenological structure (G). It fixes what the dynamics of Ψ must do, without deriving the term $x + x^2$ nor κ. Claims of a detection of the external field effect in rotation curves exist [11] and remain debated.

6 Galaxies and groups

In galaxies and groups, the low-acceleration response of Omega alone describes the data: no additional mass is required. This is the strongest observational point of the framework. One caveat is declared: the rhythm of Ψ, which clusters in the linear regime (Section 9), should be found in galaxies as well, and Omega must still explain why it is not captured there.

6.1 SPARC and the radial acceleration relation

- Data: 2 696 points from 153 galaxies of the SPARC sample [34] (standard cuts). The rotation accelerations V^2/R are measured directly (O); the baryonic mass comes from light and gas with a Newtonian potential and a stellar mass-to-light ratio (M) (Newton only, no GR).
- With the McGaugh function: $a_0 = 1.161 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, scatter 0.133 dex. The Omega regime alone fits the data with a bias of -0.004 dex.
- With the Milgrom form $g^2 = g_\mathrm{N}^2 + 2Ag_\mathrm{N}$: $A = 7.4 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, scatter 0.135 dex.
- The series mechanism with $p = 1$ gives 0.137 dex (Table 7).
- Weak lensing: lensing follows dynamics at all accelerations (Section 4.2). KiDS [10] shows a relation consistent with that of the velocities, with flat curves up to about 1 Mpc around isolated galaxies [47].

6.2 a_0 measured with less dependence on stars

SPARC sample	galaxies	a_0 (10^{-10})	effect of ± 0.1 dex on M/L	statistical	$a_0/(cH_\infty)$
All	147	1.16	0.87–1.53	—	0.213 [0.16; 0.28]
Gas $\geq 50\,\%$	60	1.12	0.96–1.29	± 0.13	$0.206 \pm 0.024 \pm 0.03$
Gas $\geq 70\,\%$	30	0.89	0.80–0.99	± 0.17	$0.164 \pm 0.032 \pm 0.018$

Table 9: a_0 from SPARC subsamples (a_0 in $\mathrm{m\,s^{-2}}$).

Gas reduces the sensitivity to the mass-to-light ratio from $\pm 30\,\%$ to $\pm 11\,\%$. The downward shift in the most gas-rich galaxies (1.6σ) is not established, since the samples are correlated; it

may come from the helium factor, uncounted molecular gas, or the distances and inclinations of dwarfs.

6.3 Distant galaxies: a_0 does not follow $H(z)$

The RC100 sample [49] contains 100 star-forming galaxies at $z = 0.6$–2.5 (M). With the present a_0 fixed by SPARC:

Hypothesis	$\Delta\chi^2$ relative to constant a_0
$a_0 \propto H(z)$	$+83$
$a_0 \propto (1+z)^{1.5}$	$+196$

- Cross-test with $a_0 = \kappa c \Gamma$ and $\Gamma \propto a^{-s}$, hence $a_0 \propto (1+z)^s$: RC100 prefers $s = +0.05\,[0; +0.15]$; with the SPARC systematic of ± 0.08 dex, $s = -0.10\,[-0.25; +0.10]$. The expansion alone gives $s = -0.03 \pm 0.03$. The two are consistent.
- With the present a_0 left free, RC100 alone prefers $s \approx -0.7$: this is misleading and must not be used.
- With the quoted intervals, $a_0(z=2)/a_0(0) \in [0.76; 1.18]$ depending on the treatment of systematics.
- Verlinde's emergent gravity gives $a_0 \approx cH_0/6$ [61]; in readings where this a_0 follows $H(z)$, it is rejected by RC100 at $\Delta\chi^2 = +83$. On this point the two theories are distinguished by observation.

6.4 Groups of galaxies [d]

In the deep low-acceleration regime, an isothermal, isotropic self-gravitating system in equilibrium satisfies

$$M\,G\,a_0 = \frac{81}{4}\,\sigma^4, \tag{15}$$

the dispersion analogue of Eq. (12). The data of Sadhu and Tian [55] (13 groups and 6 clusters, with baryonic mass and velocity dispersion) give the deviation Δ from this relation:

Systems	σ	Δ (dex)	missing mass in the Omega regime
Relaxed groups (8)	76–181 km/s	$+0.46 \pm 0.13$	none
ACO1991, ACO383	427–510 km/s	-0.34 to -0.43	$\times$2–3
Massive clusters (4)	631–901 km/s	-1.05 to -1.47	$\times$11–29

- Groups have no missing mass. The slight over-prediction is plausibly the external field effect or the dispersion being measured in the outskirts (with the total dispersion, the authors find that groups follow MOND).
- A continuous evolution ($\chi^2 = 6.7$, 2 parameters, $M_{\rm bar} \propto \sigma^{1.9}$) is preferred to a sharp transition ($\chi^2 = 13.7$, 3 parameters). The missing fraction therefore scales as $\sigma^{2.1}$.
- An ordinary superfluid would give a fraction of excitations in T^4, that is σ^8: over a factor 6 in σ, a factor 2 million in missing mass, against about 30 observed. This version is disfavoured.
- Caveats: missing gas for 5 groups, outer dispersions, different radii for groups and clusters.

6.5 Dwarf galaxies and wide binaries

- Wide binaries are compatible with the memory window τ.
- Classical dwarf spheroidals pass.
- Ultra-faint dwarfs fail for every model in which the gravity excess is tied to the baryons. This is a known difficulty of the MOND family and remains open in Omega.

7 Galaxy clusters and the boundary effect

In clusters, the low-acceleration response alone does not account for the whole acceleration. The missing part is largest in the core (lensing), and the cumulative missing mass of X-COP peaks between R_{2500} and R_{500} and then decreases. This is the known cluster problem of the MOND family. Omega addresses it with two ingredients, the boundary effect and the capture of the Ψ component; the second is required but its mechanism is not yet derived.

7.1 What X-COP requires

Five X-COP clusters [18]. The acceleration is measured from the pressure gradient of the gas; it assumes no theory of gravity, but hydrostatic equilibrium is a hypothesis (M).

Radius	baryonic acceleration	Omega alone: predicted/observed	required addition ($\times$ baryonic mass)
R_{2500} ($\approx 0.5\,\mathrm{Mpc}$)	$\approx 0.12\,a_0$	0.41	3.4 ± 0.7
R_{500} ($\approx 1.2\,\mathrm{Mpc}$)	$\approx 0.07\,a_0$	0.68	1.0 ± 0.1
R_{200} ($\approx 1.8\,\mathrm{Mpc}$)	$\approx 0.05\,a_0$	0.95	0.1 ± 0.2

- The central deficit is the known deficit of MOND, not specific to Omega.
- The cumulative missing mass peaks between R_{2500} and R_{500} (about 0.27, 0.33 and 0.10 M_{500} at R_{2500}, R_{500}, R_{200}) and then decreases. A decreasing cumulative mass implies a negative missing density beyond R_{500}: either a non-thermal pressure support of 10–25 % (which X-COP itself suggests at R_{200}), or a real effect by which Omega predicts too much acceleration outside the boundary.
- Lensing of CLASH clusters [21] shows a missing density that is maximal and flat in the core, where $g \gg a_0$. There Omega is in its local-GR regime, and the excess requires something that gravitates.

7.2 The boundary effect (g)

Each system has a surface on which the observed acceleration equals a_0. The hypothesis is that forces concentrate on this boundary and that their intensity decides the type of structure. Galaxies and groups also have a boundary, with no excess: the effect must depend on a property present in clusters. In the data of Sadhu and Tian, stars make up a median 92 % of the baryons of groups but only 14 % of those of clusters; the rest is very hot gas under strong pressure. This points to a boundary force tied to the pressure or the thermal energy of the matter that crosses the boundary.

The present form of the hypothesis is:

> **P5 bis.** In a cluster, the mass excess is confined inside the boundary and dies out where the observed acceleration falls to a_0. The cut-off radius must follow, cluster by cluster, the radius where $g = a_0$, and not a fixed fraction of an overdensity radius.

For an isothermal sphere the radius where $g = a_0$ is $2kT/(\mu_{\rm mol} m_p a_0)$, so it scales as T, whereas self-similar scaling gives $T^{1/2}/E(z)$. A first test on 16 CLASH clusters (scale radius of the missing mass, $z = 0.21$–0.69, $kT = 5.8$–$15.5\,\mathrm{keV}$) is not informative: the residual scatter is similar for the boundary law, the self-similar law and a constant (0.22, 0.25 and 0.28 in $\ln r_s$ on the 8 relaxed clusters). The temperature lever arm is too short. The median scale radius is 0.7 times the radius where $g = a_0$. Since g at radii defined by an overdensity varies as $M^{1/3}$, "$g \approx a_0$ at R_{2500}" is partly a consequence of how these radii are defined; a decisive test needs clusters spanning at least a factor 10 in mass.

7.3 The mass term of Ψ in clusters

In the static limit, Ψ acquires a mass term μ (Section 9.5). Following the analysis of AeST clusters [16], it gives an acceleration first above the low-acceleration value, then below it (negative phantom mass), then oscillations. This is qualitatively the X-COP pattern of a cumulative missing mass that peaks and then decreases. The radius where the regime changes is

$$r_C \approx \tfrac{1}{3}\left(\frac{18\, r_M}{\mu^2}\right)^{1/3}, \qquad r_M = \sqrt{\frac{GM}{a_0}}. \tag{16}$$

- If $\mu = \Gamma/c$ ($\mu^{-1} = 1.78\,\mathrm{Gpc}$), $r_C = 132\,\mathrm{Mpc}$ for a cluster of $10^{15} M_\odot$: no effect in clusters. This link is excluded.
- The X-COP pattern near 1 Mpc requires $\mu^{-1} \approx 1\,\mathrm{Mpc}$ ($r_C = 0.9\,\mathrm{Mpc}$ for $10^{15} M_\odot$), just at the limit tolerated by galaxies ($r_C = 0.19\,\mathrm{Mpc}$ for $10^{11} M_\odot$).
- The mass term acts as an effective mass $(\mu^2/G)\int |\Phi| r^2 dr$, which grows as r^3 and vanishes at the centre. It cannot produce the core excess; at best it shapes the periphery.

7.4 The core requires captured Ψ

The core of clusters therefore requires that part of the pressureless component carried by the rhythm of Ψ be captured: about 1 to 3.4 times the baryonic mass, below the cosmic reservoir of 5.4. Capture does not depend on the acceleration (galaxy centres also have $g > a_0$, without excess) but on the depth of the well: nothing below $\sigma \approx 200\,\mathrm{km/s}$, an increasing capture above, with a smooth law (missing fraction $\propto \sigma^{2.1}$).

Results obtained so far on the capture mechanism:

- No scale of the present model gives a velocity of a few hundred km/s: $\sqrt{a_0/\mu} = 1\,900\,\mathrm{km/s}$ for $\mu^{-1} = 1\,\mathrm{Mpc}$; $a_0/\Gamma = 21\,000\,\mathrm{km/s}$; $a_0\tau = 0.5$–$1.8\,\mathrm{km/s}$.
- A capture proportional to the shift of the rhythm of Ψ at the bottom of the well ($\propto |\Phi|/c^2$), without threshold, reproduces the slope $\propto \sigma^2$ but predicts an outer slope of the missing density between 0 and -1, whereas CLASH lensing gives a slope steeper than -3.7. It is excluded.
- A capture depending on the dynamical history (disturbed versus relaxed clusters) is not supported: 8 disturbed and 8 relaxed CLASH clusters show the same excess ($\Delta \log \eta = +0.01 \pm 0.09$ at controlled temperature).

The capture law is the main open problem of the cluster sector. Three directions remain: a new scale of the condensate giving a few hundred km/s; a scale-free capture law producing a fraction in $\sigma^{2.1}$; or a dependence on the formation history over billions of years.

7.5 The Bullet Cluster

In the Bullet Cluster the lensing mass follows the galaxies, not the gas. The pressureless component of Ψ has a negligible sound speed in the present architecture: two clouds of it crossing at 4 700 km/s behave as collisionless (a coherent field superposes by interference; at large Mach number the clouds cross). The captured component therefore follows the galaxies, as observed. The memory τ plays no role here: its wake is 0.7–2.4 kpc, far below the observed offset of 150–200 kpc.

7.6 Anisotropy of cluster scaling relations

For 313 clusters of Migkas et al. [43], the L_X–T relation with a dipolar normalisation scanned over the sky gives a dipole of 0.038 ± 0.027 dex (1.4σ), lowest towards $(l, b) \approx (260^\circ, -30^\circ)$, close to the Migkas region. Pantheon+ supernovae at $z < 0.1$ in this direction show a distance variation of -0.4 ± 0.6 %, whereas a "distance" reading would require about 4 %. If the anisotropy is real, it lies in the clusters, not in space. It is documented as a property of clusters; it is not a test of the primary wave, whose imprint precedes all objects.

8 Creation and expansion without dark energy

The accelerated expansion is a motion of the vacuum of the condensate, its dilation, at zero total energy. Braking by matter comes from an energy balance. The fit to distances is, by construction, identical to that of flat ΛCDM: the difference is one of interpretation, and it becomes observable only through the link between the expansion and a_0 (Section 8.6).

8.1 What distances measure

Supernovae and baryon acoustic oscillations measure distances as functions of redshift, that is, an expansion history $H(z)$. They do not measure the nature of what drives it. A reading without dark energy in which time does not flow at the same rate everywhere (the timescape model [69]) fits Pantheon+ [57] with $\Delta\chi^2 = +5.1$ relative to ΛCDM (+2.1 to +7.1 depending on the low-redshift cut, 1.4 to 2.6σ); only a matter-only universe with a single clock is excluded massively (+698). In Omega, such an effect could not come from bound regions (clock offsets $1 + \Phi/c^2 \approx 10^{-5}$); it is not used.

8.2 Emergent cosmology (g)

If the condensate is created at the rate Γ, its density stays constant while it expands, and the scale factor grows as $\exp(\Gamma t/3)$: accelerated expansion with asymptotic rate $H_\infty = \Gamma/3$. The acceleration is kinematic. With creation, $d\ln\rho/dt = \Gamma - 3H$: the density is constant only if $H = \Gamma/3$ at all epochs. In the acoustic version, a constant sound speed would make the emergent scale factor evolve at the rate $(\Gamma - H)/2$, and the emergent universe would contract during the matter era; this version is tenable only if ρ/c_s is constant (equation of state $P \propto \rho^3$), a hidden assumption. The tensor version below removes it.

What is created is the medium, not baryons and not photons. Creating baryons at the rate needed to sustain the expansion would require a baryon density about 1.4 times the critical density, excluded by fast radio bursts ($\Omega_b \approx 0.05$ [41]) by a factor 30; creating photons would produce 3×10^4 times the CMB energy. Condition (G): the energy of a condensate at equilibrium does not gravitate for its own excitations (Volovik's argument [63]), not yet proven.

8.3 The rolling vacuum [d]

On a flat background with universal time, the spatial part of the tensor is $C_{ij} = A(t)\,\delta_{ij}$ and the emergent spatial metric is $a^2 \propto A$. With the kinetic term $\frac{3}{2}\dot{A}^2$ and the creation part of the potential (2), the potential is inverted along the single dilation direction, while the shears (the gravitons) remain stable:

$$L_A = \tfrac{3}{2}\dot{A}^2 + \tfrac{3}{2}m^2A^2 \quad \Rightarrow \quad \ddot{A} = m^2 A \quad \Rightarrow \quad A \propto e^{mt}, \quad H = \frac{\dot{a}}{a} = \frac{\dot{A}}{2A} = \frac{m}{2}. \tag{17}$$

With $H_\infty = \Gamma/3$, $m = 2\Gamma/3$.

- The energy $E_A = \frac{3}{2}(\dot{A}^2 - m^2A^2)$ vanishes on the solution: the solution is exactly de Sitter, at zero total energy. What the dilation gains in kinetic energy, the potential loses. This is permanent creation in the strict sense, without any energy input and without a dark-energy fluid. $\hbar m \approx 2.4 \times 10^{-33}$ eV.
- Attractor: for a non-zero initial energy, $A \propto \sinh(mt)$ or $\cosh(mt)$, which become exponential. The regime does not depend on initial conditions.
- Alone, this vacuum gives pure de Sitter, $\chi^2 = 2\,059.9$ on Pantheon+ against 1 402.9 for ΛCDM ($\Delta\chi^2 \approx +657$): matter must brake the dilation.

8.4 Braking by matter: energy balance [d]

Without Friedmann's equation, what remains is the conservation of the total energy of the system (dilation, created vacuum, matter). For matter to brake, the dilation energy must be negative and proportional to $-a\dot{a}^2$: it is the "gravitational" energy of the expansion itself. The balance gives

$$H^2 \propto \rho_{\text{matter}} + \rho_{\text{vac}} + \frac{E}{a^3}. \tag{18}$$

- $E = 0$: exactly the distances of ΛCDM; creation happens at zero total energy (the negative dilation energy compensates that of the created vacuum). Precedent: the zero-energy universe of Tryon [59].
- $E \neq 0$: an a^{-3} term.
- In the tensor version, the statement "same history as ΛCDM" assumes that the spatial eigenvalues obey this law of braking plus creation: hypothesis (G), not yet derived from the full kinetic term of the tensor (Section 16).

Fit to DESI DR2 BAO [15], the CMB acoustic angle and Pantheon+:

$$\Gamma = 5.46 \times 10^{-18}\ \text{s}^{-1}, \qquad 1/\Gamma \approx 5.8\ \text{Gyr}, \qquad 1/H_\infty \approx 17.4\ \text{Gyr}, \tag{19}$$

with $\chi^2 = 1\,430.07$, identical to ΛCDM. These are characteristic times, not ages. A creation rate that varies smoothly gives $w \approx -1.02$ at best fit; the preference of DESI for a crossing of $w = -1$ in the past ($\Delta\chi^2 \approx -16$ for the w_0–w_a parametrisation) is not reproduced (Section 10).

8.5 The link between a_0 and Γ (g)

$a_0 = \kappa c \Gamma$ and $H_\infty = \Gamma/3$ come from the same source. Measured target:

Target: $a_0/(cH_\infty) \approx 0.16$–$0.24$, that is $\kappa \approx 0.055$–0.095 depending on the shape of the transition (about ±35 %). The link $a_0 \sim c\sqrt{\Lambda}$ has been proposed in the MOND literature [45]; what is specific to Omega is that a_0 follows the creation rate. Dimensional analysis does not fix κ.

Route to H_∞	cH_∞ ($\mathrm{m\,s^{-2}}$)	$a_0/(cH_\infty)$
BAO (ΛCDM sound horizon, (M)) plus supernovae	5.45×10^{-10}	0.213
SH0ES (73.0 km/s/Mpc) plus supernova shape	5.80×10^{-10}	0.200

Mechanisms examined for κ. Each mechanism below was written before the computation, without factors chosen afterwards; factors such as $1/(2\pi)$, $2/(3\pi)$, $1/(4\pi)$ all fall in the target interval and are excluded as numerology.

Mechanism	law	$a_0/(cH_\infty)$	verdict
Inertia from the excess temperature (Milgrom 1999 [45]), transposed	$A = cH_\infty$	1	too large, $\times$7.4–7.8
Equal creation and acceleration temperatures	$a_0 = cH_\infty$	1	too large, $\times$5
The field establishes itself as fast as the vacuum dilates	$g/c = H_\infty$	1	too large, $\times$3.5–4.7
The GR region blocks the unstable dilation mode (rate $m = 2H_\infty$) over the length c^2/g	$g = cm/\pi$	0.64	too large, $\times$2.2–3.0
Field energy equals the dilation energy of the vacuum	$g^2 = 3c^2H_\infty^2$	1.73	too large, $\times$6–8
Emergent gravity of Verlinde with H_∞ (imported)	$g = cH_\infty/6$	0.167	borderline; foreign to Omega

Table 10: Mechanisms linking a_0 to the creation rate.

Every mechanism that compares the field directly with the creation rate gives a number of order one, whereas the data require about 0.2. The missing reduction factor must come from a structure absent from the present model: a counting of degrees of freedom of the condensate, a geometry of the boundary, or a ratio between two scales of the condensate. κ **is not derived.**

8.6 Which rate does a_0 follow? (g)

The scale a_0 appears in the energy of Ψ. A local energy can depend only on local quantities. Three sources are possible:

Source of a_0	Behaviour	Verdict
(a) A constant of the action, simply equal to $\kappa c\, m$	a_0 strictly constant	possible, but the link a_0–Γ becomes the unexplained equality of two constants: the "same source" is only a coincidence
(b) The local expansion rate of the clock, $\theta = \nabla_\mu u^\mu$	contains the braking by matter: $a_0 \propto H(z)$	excluded by RC100 ($\Delta\chi^2 = +83$)
(c) The vacuum state of the condensate, $\lvert V(A)\rvert/A^2 = \frac{3}{2}(m^2 - \lambda A^2)$	$a_0 \propto H_{\mathrm{vac}}$, the dilation rate of the vacuum alone, insensitive to matter	the only source that realises "same source" structurally

With (c), at zero total energy of the vacuum, the kinetic energy of dilation equals $-V(A)$: the local quantity $\sqrt{|V(A)|}/A$ is exactly the dilation rate of the vacuum, $\dot{A}/A = \sqrt{m^2 - \lambda A^2}$. a_0 then does not need to be a second constant. Consequences:

- a_0 tends to zero at the turnaround of the cycle, when creation stops: the boundary effect disappears with the creation that produces it.
- If the turnaround term is absent ($\lambda = 0$), (a) and (c) coincide: a_0 is constant and $w = -1$.
- If $\lambda \neq 0$, the evolution of a_0 is tied without free parameter to the equation of state of the vacuum (Section 10.5).

Status: structural argument (G), not a derivation. It does not fix κ; it depends on $a^2 \propto A$ and on the zero total energy of the vacuum.

9 The field Ψ: pressureless component and co-creation

9.1 Why a pressureless component is required

- The height of the third acoustic peak of the CMB requires a non-baryonic gravitational effect at recombination, equivalent to about 5.4 times the baryon density ($\omega_c/\omega_b = 5.364 \pm 0.065$ (M) [51]). At recombination $g \approx 14\,a_0$: Omega is in its local-GR regime there, and the standard analysis of the peaks is legitimate within the hierarchy of Omega. Baryons and photons alone do not reproduce this peak.
- The cores of clusters require a centred addition (Section 7).
- Rule: nothing is added by hand. The component must follow from the dynamics of Omega.

9.2 Theorem: one clock cannot give both the pressureless component and the low-acceleration response [d]

When time is universal and is no longer a free coordinate at each point, the equations lose a local constraint and a dust-like term appears automatically (Mukohyama [48] for Hořava gravity; Chamseddine and Mukhanov [12] for mimetic gravity). It is not made of particles, and its amount is an integration constant.

This geometric dust appears when the clock follows the gradient of a normalised phase, $u_\mu = \partial_\mu T$ with $u^\mu u_\mu = -1$. Its acceleration then vanishes identically:

$$a_\mu = u^\nu \nabla_\nu u_\mu = u^\nu \nabla_\nu \nabla_\mu T = u^\nu \nabla_\mu \nabla_\nu T = u^\nu \nabla_\mu u_\nu = \tfrac{1}{2}\,\partial_\mu (u^\nu u_\nu) = 0. \tag{20}$$

A low-acceleration response carried by the acceleration of the clock (a function $f(a)$) is then impossible. A clock without acceleration allows the dust but forbids the response; an accelerated clock allows the response but forbids the geometric dust. In addition, Hořava-type versions with uniform rate have an unstable or ill-defined scalar mode [7]. **Two fields are needed**: in the present architecture, Ψ carries both, through two different variables, its gradient Y and its rhythm Q.

9.3 Dynamics of the rhythm [d]

Let $Q = d\Psi/dt$ be the rhythm of Ψ measured by the universal clock and $K(Q)$ its energy function. On a homogeneous background, the action is invariant under $\Psi \to \Psi + \text{const}$, and the associated charge

$$\Pi = K'(Q) \qquad \text{obeys} \qquad \frac{d}{dt}\left(a^3 \Pi\right) = 0. \tag{21}$$

The energy density is the Legendre transform,

$$\rho_\Psi = Q\,\Pi - K(Q). \tag{22}$$

If K has an extremum at $Q_0 \neq 0$ and Q stays close to Q_0, then $\rho_\Psi \simeq \text{const} + Q_0\,\Pi \propto a^{-3}$: a pressureless component carried by a coherent field.

9.4 Co-creation (g)

The creation that produced matter at the bounce of the cycle produced at the same time charge of Ψ. If n_c units of charge are created per baryon,

$$\frac{\rho_\Psi}{\rho_b} = n_c\,\frac{\hbar Q_0}{m_p c^2}. \tag{23}$$

The bounce can fix n_c (a counting); it cannot fix Q_0, which is the equilibrium rhythm of Ψ, a property of $K(Q)$ and not of the creation event.

Principle. The charge of Ψ is the baryon charge itself, carried in condensate form. Then $n_c = 1$ exactly, with no parameter, and

$$\hbar Q_0 = 5.364\, m_p c^2 = 5.03 \pm 0.06 \text{ GeV}. \tag{24}$$

The free parameter changes nature: it is no longer an adjusted amount of dust, it is the energy of the proper rhythm of Ψ, a frequency of about 10^{24} Hz. For a phase that carries a conserved charge, the rhythm is the chemical potential of that charge: $\hbar Q_0$ is the energy it costs to add one unit of charge to the condensate.

Consequences of the principle:

- the fluctuations of Ψ and of the baryons are adiabatic everywhere (no isocurvature, as Planck requires);
- the total charge of the two poles of the battery is zero (as many antibaryons in the twin as baryons here; Section 11);
- baryon number is conserved globally. Our excess of baryons is a separation of charges between the poles, not a net creation, so the Sakharov conditions, which apply to a closed universe, are met at the level of the pair. An out-of-equilibrium process at the bounce exists; the "electromotive force" that sends the positive charges to our pole remains to be derived.

Chronology: fast radio bursts exclude a substantial late creation of baryons; co-creation took place during the recreation of the cycle, before recombination. Two processes therefore coexist: continuous creation of space at the rate Γ, and creation of matter and of charge of Ψ at the bounce.

Equilibrium laws are excluded [d]. If the charge of Ψ is in chemical equilibrium with nucleons, the chemical potentials are equal, $\hbar Q_0 = m_p c^2$ and $\rho_\Psi/\rho_b = 1$, against 5.364 ± 0.065 measured: excluded (67 times the uncertainty). At the quark stage, before confinement, the baryon chemical potential is about η times the temperature, which would make $\hbar Q_0$ billions of times too small: excluded. The ratio 5.36 is therefore not an equilibrium value but a frozen proper scale of Ψ. Since $\hbar Q_0 > m_p c^2$, a unit of charge in the condensate would gain about 4.1 GeV by turning into a nucleon: the principle requires a barrier (topological or kinetic) that forbids this conversion after the bounce; otherwise the component would have turned into matter and radiation. This is consistent with the absence of any non-gravitational signal (prediction P6) but must be derived.

Related constructions. Structurally, co-creation resembles asymmetric dark matter [29], which has a particle near 5 GeV; the difference is that here the charge belongs to a coherent field. A coherently oscillating scalar also behaves on average as dust ("wave dark matter"), with a bound $m \gtrsim 2 \times 10^{-20}$ eV from the Lyman-α forest [54]; Omega's component is not of that type (its rhythm is a chemical potential, not a mass).

9.5 Stiffness of Ψ, the mass term and the linearity condition [d]

In a static potential well Φ, the rhythm of Ψ read by the clock shifts by $\delta Q = -Q_0\Phi/c^2$, and this shift costs energy through K. This generates a mass term in the equation for the potential. With the susceptibility $\chi = K''$ (charge added per unit of rhythm), the energy shift sources gravity as $\delta\rho\, c^2 = Q_0\, \delta\Pi = Q_0\chi\, \delta Q$, and the Poisson equation acquires a term $-\mu^2\Phi$ with

$$\mu^2 c^2 = 4\pi G\, \chi\, Q_0^2 \tag{25}$$

(in units where χQ_0^2 is a mass density per unit potential). For a power law $K \propto |Q - Q_0|^n$, $\Pi = nA\, \delta Q^{n-1}$ and $\chi = (n-1)\Pi/\delta Q$; combining with $\rho_\Psi c^2 = Q_0\Pi$,

$$\boxed{\frac{\delta Q}{Q_0} = (n-1)\,\frac{4\pi G\,\rho_\Psi}{\mu^2 c^2}}, \qquad \rho_\Psi = \rho_{\rm dust}\left[1 + \frac{n-1}{n}\,\frac{\delta Q}{Q_0}\right]. \tag{26}$$

Up to conventions, this is the relation $\rho = m^2(\delta Q/Q_0)/(4\pi G f_G)$ published for AeST [46]. Ψ behaves as dust only if $\delta Q \ll Q_0$.

Three consequences:

1. $\hbar Q_0$ **drops out.** Co-creation (5.03 GeV) fixes neither μ nor the stiffness. If 5 GeV were the only scale of Ψ (the normalisation of ghost condensation [2]), one would have $\mu^{-1} \approx M_{\rm Pl}/M^2 \approx 27$ m, excluded by galaxies ($\mu^{-1} \gtrsim 1$ Mpc). Ψ has at least two scales.
2. **Constant stiffness is excluded.** For $n = 2$ and $\mu^{-1} = 1$ Mpc, $\delta Q/Q_0 = 2 \times 10^{-8}$ today, 26 at recombination and 790 at matter–radiation equality: Ψ would not be dust when the acoustic peaks form. Limiting the excess to 10 % would require $\mu^{-1} \leq 88$ kpc (recombination) or ≤ 16 kpc (equality), a factor 60 below what galaxies require.
3. **The stiffness must grow with the density of Ψ.** For $K \propto |Q - Q_0|^n$, $\mu \propto \rho_\Psi^p$ with $p = (n-2)/[2(n-1)]$. With $\mu^{-1} = 1$ Mpc today at mean density:

n	p	excess at recombination	excess at equality	extra energy / radiation at nucleosynthesis
2	0	13	394	5×10^{12}
2.6	0.19	1 %	8 %	1 900
3	0.25	0.1 %	0.5 %	1.5
3.2	0.27	0.04 %	0.2 %	0.12
3.5	0.30	0.02 %	0.06 %	0.005

Table 11: Linearity of the Ψ component for power-law stiffness.

Dust at equality requires $p \geq 0.18$; nucleosynthesis (extra energy below 5 % of the radiation) requires $p \geq 0.28$. The branch $n < 1$ ($p > 1/2$), in which $\delta Q/Q_0$ decreases towards the past, also passes. AeST reaches the same conclusion with stiffening functions $K = 2K_2 Z_0^2[\cosh((Q - Q_0)/Z_0) - 1]$ or exponential forms, with $Z_0/Q_0 \sim 10^{-9}$ to 10^{-17} [58]; Omega adopts this class. For the cosh form, $\mu^2 \propto \cosh x$ with $\sinh x \propto \rho_\Psi$: beyond the transition, $\mu^{-1} \propto (1+z)^{-3/2}$.

This is a derived form of "each body has its own rules": the range of the mass term depends on the local density of Ψ. It is shorter in clusters than in galaxies and cannot increase with redshift; its amount of variation at low redshift depends on the present position $x_0 = \delta Q/Z_0$, which is free (μ^{-1} reduced by 0.40 at $z = 1$ if $x_0 = 1$, by 0.89 if $x_0 = 0.1$).

9.6 Linear growth and the Lyman-α forest

In the linear cosmological regime, the rhythm of Ψ clusters like dust without appreciable sound speed, and the low-acceleration term acts only in the quasi-static non-linear regime. The linear spectrum at $z = 3$ is then that of the reference computation. DESI DR1 [13] measures, at the pivot $k_* = 0.009\,\mathrm{s/km}$ and $z_* = 3$, $\Delta^2_* = 0.379 \pm 0.032$ and $n_* = -2.309 \pm 0.019$; Omega gives 0.376 and -2.299 ($\chi^2 \approx 0.3$ for 2 data). This agreement is inherited: at linear cosmological scales, the Ψ component gravitates exactly as cold dark matter. Non-linear growth in this framework has not been simulated, in Omega or in AeST.

A pressureless component with a constant sound speed is excluded: galaxies require $c_s \gtrsim 150$–$190\,\mathrm{km/s}$ today, whereas DESI requires $c_s \lesssim 50\,\mathrm{km/s}$ at $z \approx 3$ (at $300\,\mathrm{km/s}$ the slope n_* falls to -4.5). The present architecture avoids this conflict because the sound speed of the rhythm is negligible.

9.7 Galaxies: no halo

In galaxies, in the quasi-static regime, Ψ is in its low-acceleration regime and does not form a halo of particles; lensing equals dynamics ($\Phi = \Psi$ in the quasi-static limit of AeST [62]). SPARC tolerates at most about one baryonic mass of clustered dust: with a minimal halo of 5.4 times the baryons added to the low-acceleration effect, the bias is $-0.112\,\mathrm{dex}$; with a realistic halo, $-0.231\,\mathrm{dex}$, against $-0.004\,\mathrm{dex}$ for the low-acceleration effect alone.

9.8 The central tension

In the linear regime, the rhythm of Ψ clusters down to galactic scales (as the Lyman-α forest requires). Galaxies form from these overdensities and should contain it; SPARC tolerates about one baryonic mass of it, while clusters require 1 to 3.4. AeST assumes that Ψ is at equilibrium in quasi-static systems without demonstrating it. Omega must provide a mechanism that distributes this component according to the system. The density dependence of the mass term (Section 9.5) is a first ingredient of such a mechanism; the capture law itself is not derived.

10 The cycle: turnaround, compression and bounce

Creation is eternal and discontinuous: a sequence of episodes with neither a first nor a last. Each episode is an expansion, a turnaround, a compression and a bounce. The solution is a periodic elliptic function of the universal time.

10.1 The turnaround [d]

If the dilation potential is bounded by a quartic term, a turnaround follows. At zero total energy,

$$\dot{A}^2 = m^2 A^2 - \lambda A^4 \quad \Rightarrow \quad A(t) = A_{\max}\,\mathrm{sech}(mt), \qquad A^2_{\max} = \frac{m^2}{\lambda}. \tag{27}$$

Indeed $\dot{A} = -mA_{\max}\mathrm{sech}(mt)\tanh(mt)$, so $\dot{A}^2 = m^2A^2(1 - \mathrm{sech}^2) = m^2A^2 - \lambda A^4$. With $a^2 \propto A$, the dilation rate of the vacuum is

$$H_{\mathrm{vac}} = \frac{\dot{A}}{2A} = H_\infty\sqrt{1 - \left(\frac{a}{a_{\max}}\right)^4}, \qquad \rho_{\mathrm{vac}} \propto 1 - x^4, \quad x = \frac{a}{a_{\max}}. \tag{28}$$

From $d\ln\rho_{\rm vac}/d\ln a = -3(1+w)$:

$$1+w = \frac{4}{3}\frac{x^4}{1-x^4} \geq 0. \tag{29}$$

Signature today: the vacuum energy decreases slightly, $w > -1$ and increasing. The model always has $w \geq -1$: it cannot reproduce a crossing of $w = -1$.

10.2 Distinct episodes: the bounce [d] [n]

The sech solution takes an infinite time to return towards $A \to 0$: it would be a single infinite episode. A third term gives the stiffness of the strongly compressed condensate:

Term	Role
$-\frac{3}{2}m^2A^2$	creation, the rolling vacuum
$+\frac{3}{2}\lambda A^4$	the turnaround
$+\frac{3}{2}\varepsilon/A^2$	the bounce, at the microscopic scale of the condensate

At zero energy, $\dot{A}^2 = m^2A^2 - \lambda A^4 - \varepsilon/A^2$. With $u = A^2$, $\dot{u} = 2A\dot{A}$:

$$\dot{u}^2 = 4\left(m^2u^2 - \lambda u^3 - \varepsilon\right). \tag{30}$$

The right-hand side is a cubic in u: the solution is a periodic elliptic function. A oscillates indefinitely between a minimum (bounce) and a maximum (turnaround).

- Direct integration of the equation of motion: successive minima are identical (0.031631 at each bounce in reduced units). There is neither a first nor a last episode.
- Signature of the bounce term: a correction in $-1/a^8$ to the expansion rate, which dominates radiation ($1/a^4$) at strong compression and ensures the bounce before any singularity. Loop quantum cosmology obtains the same law, with a correction $-\rho^2/\rho_c$ [3]. In Omega, the critical density is that of the condensate, at least $(2\times10^{16}\,\text{GeV})^4$ from the Crab bound, far above the thermalisation threshold of the CMB.
- About **67 e-folds** separate the bounce from the turnaround (bounce at the scale of the condensate, turnaround near $a_{\max} \approx 2.5$). This is the order of magnitude that inflation has to assume; here it follows from the ratio between the microscopic scale and the creation rate. The stiffness ε must be very small compared with the creation scale: the same hierarchy as that required by the Crab.
- Limitation: this model contains only the vacuum. Near the bounce, matter and radiation dominate the budget; it remains to be checked, with all components, that the stiffness of the condensate dominates them.

10.3 Cycle and battery

When the potential between the two poles of the battery (Section 11) equalises, creation stops, the expansion reverses and compression begins, during which the CMB and matter are recreated. Battery and cycle can then be the same physics. If the poles swap from one episode to the next, our universe and the twin exchange their roles at each cycle, like an alternating current: creation is eternal without exhausting the reservoir. Two laws are therefore considered for the vacuum energy, each with a single parameter $x_0 = 1/a_{\max}$:

- **L-ret** (quartic potential): $\rho_{\rm vac} \propto 1-x^4$, $\Gamma \propto H_{\rm vac} \propto \sqrt{1-x^4}$;
- **L-bat** (linear discharge until the poles are equal at $a_{\rm eq}$): $\Gamma \propto 1-(a/a_{\rm eq})^3$, $\rho_{\rm vac} \propto \left(1-(a/a_{\rm eq})^3\right)^2$, hence $1+w = 2x^3/(1-x^3)$.

10.4 Joint fit [n]

Fit to DESI DR2 BAO, the CMB acoustic angle and Pantheon+ (one parameter more than ΛCDM):

Law	vacuum energy	best $\Delta\chi^2$ vs ΛCDM	2σ bound	s today at the bound	$1+w_0$ at the bound
L-ret	$1-(a/a_{\max})^4$	-0.4 ($a_{\max}\approx 2.5$)	$a_{\max}\geq 1.81$	0.21	0.14
L-bat	$(1-(a/a_{\rm eq})^3)^2$	-0.2 ($a_{\rm eq}\approx 5$)	$a_{\rm eq}\geq 2.74$	0.15	0.10

Table 12: Cycle and battery laws against the expansion data. $s=-d\ln\Gamma/d\ln a$.

Both laws are compatible with the data without being preferred (the w_0–w_a parametrisation gains -15.8 thanks to a crossing of $w=-1$ that Omega forbids). Profile of χ^2 for L-ret: $\Delta\chi^2=-0.05,-0.23,-0.41,+0.72,+3.18,+8.74$ for $a_{\max}=5,3.33,2.5,2.0,1.82,1.67$. The turnaround would occur at the earliest some ten to fifteen billion years from now; it is a duration towards the future, not an age.

Two consequences:

- **The a^4 shape is not discriminating in practice.** At the bound, $1+w=0.008$ (L-ret) or 0.012 (L-bat) at $z=1$; the difference from a thawing quintessence ($1+w\propto a^3$) is about 0.005, below the precision of surveys (0.03–0.05 per bin). The real test is the absence of a crossing of $w=-1$ (prediction P2).
- **Open coherence point.** The fit constrains only where the vacuum energy vanishes. With braking by matter added in H^2, the emergent expansion does not reverse by itself at $a_{\max}$; with $a^2\propto A$ strictly, it reverses but matter no longer brakes. The moment of the turnaround depends on the braking law, not yet derived.

10.5 a_0 follows the vacuum: a parameter-free relation (g)

If a_0 follows the dilation rate of the vacuum (Section 8.6), $a_0\propto\Gamma\propto H_{\rm vac}$. Its value in the distant past ($x\to 0$) relative to today is then fixed by the present equation of state.

L-ret. $a_0(z\gg 0)/a_0(0)=1/\sqrt{1-x_0^4}$. From Eq. (29), $x_0^4/(1-x_0^4)=\frac{3}{4}(1+w_0)$, so $1/(1-x_0^4)=1+\frac{3}{4}(1+w_0)$.

L-bat. $a_0(z\gg 0)/a_0(0)=1/(1-x_0^3)$ and $1+w_0=2x_0^3/(1-x_0^3)$, so $1/(1-x_0^3)=1+\frac{1}{2}(1+w_0)$.

$$\boxed{\frac{a_0(z\gtrsim 1)}{a_0(0)}=\sqrt{1+\tfrac{3}{4}\,(1+w_0)}\ \ \text{(L-ret)},\qquad 1+\tfrac{1}{2}\,(1+w_0)\ \ \text{(L-bat)}} \tag{31}$$

The excess of a_0 in the past equals 0.37 to 0.5 times $(1+w_0)$, without free parameter. At the present bounds: at most 5 % (L-ret: 1.041 at $z=0.5$, 1.048 at $z=1$, 1.050 at $z=2$; L-bat: 1.036, 1.045, 1.049). If a vacuum with $w_0\approx -0.75$ and no crossing of -1 were confirmed, a_0 should be 9 to 12.5 % larger at $z\gtrsim 1$.

No other theory ties the evolution of a_0 to w_0 in this way: in MOND and AeST, a_0 is constant; in Verlinde's emergent gravity it follows $H(z)$, which RC100 excludes. This is prediction P3 bis (Section 17).

10.6 What the compression phase must produce

- the CMB photons, thermalised (Section 13);

- the primordial fluctuations, with $n_s \approx 0.965$ (Section 12);
- the baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$, not derived;
- the charge of Ψ and the ratio 5.36 (Section 9).

Nucleosynthesis after the bounce. Light elements are produced after the bounce. Near 1 MeV the acceleration is far above a_0 (local-GR regime), the bounce term is at most $(1\,\mathrm{MeV}/2 \times 10^{16}\,\mathrm{GeV})^4 \approx 6 \times 10^{-78}$ of the density, and the created vacuum is negligible: the expansion is that of a radiation-dominated universe. Nucleosynthesis is therefore standard, and the agreement between ω_b from deuterium and from the CMB is automatic, provided that Ψ does not add to the radiation. If Ψ was relativistic before its change of phase (Section 12) and kept its energy at the transition, its share would be $2\,890/(1+z_t)$ of the radiation; $\Delta N_{\mathrm{eff}} < 0.3$ then imposes $1 + z_t \gtrsim 7 \times 10^4$.

11 The twin universe: the battery

11.1 Where the polarity lies (g)

- Not directly on matter and antimatter, which would annihilate on contact, whereas the two sectors do not touch.
- On a conserved charge of the condensate, that of Ψ: our universe is the positive pole, the twin the negative pole, the total charge is zero.
- Matter and antimatter follow the charge at co-creation: each baryon created here has its antibaryon in the twin. Antimatter in the twin, matter here.
- Creation is a current from the twin towards us, driven by the potential difference between the poles.
- The twin does not act on our matter through gravity; it only feeds creation. This distinguishes it from the negative-mass twin of the Janus model, and from "mirror dark matter", in which the twin is a source of hidden mass (excluded by the Bullet Cluster).

11.2 What the constancy of the creation rate implies

A battery discharges and its current falls. If the creation rate is written as a power law, $\Gamma \propto a^{-s}$ with s constant at all epochs, the expansion data give $s = -0.03 \pm 0.03$, that is $s \lesssim 0.025$ to 0.03 at 2σ:

$$\left| \frac{d \ln \Gamma}{dt} \right| \lesssim 0.03\, H_\infty. \tag{32}$$

For a discharge proportional to the content of the reservoir, this means a discharge time of at least $40/H_\infty$, about 700 Gyr ($1/H_\infty = 17.4$ Gyr measures the present expansion rate, not an age), and a reservoir holding at least about 40 times the charge transferred during one e-fold of expansion.

To keep the vacuum energy density constant while our volume grows, creation transfers $3\rho_{\mathrm{vac}} V$ per unit of $\ln a$. A reservoir of capacity Q_{res} empties at the rate $3\rho_{\mathrm{vac}}/Q_{\mathrm{res}}$ per unit of $\ln a$. Hence, at equivalent volume,

$$M_{\mathrm{twin}} \approx \frac{3\rho_{\mathrm{vac}}}{s}, \qquad \frac{M_{\mathrm{twin}}}{M_{\mathrm{matter}}} \approx \frac{2\,\Omega_{\mathrm{vac}}}{\Omega_{\mathrm{matter}}\,(1+w_0)}. \tag{33}$$

Precision on s	reservoir $\geq$	ratio to our matter	ratio to our baryons
Today ($s < 0.025$, $\approx 2\sigma$)	$83\times$ critical density	$\geq 265\times$	$\geq 1\,700\times$
Final DESI / Euclid ($s < 0.01$)	$206\times$ critical density	$\geq 660\times$	$\geq 4\,200\times$

Table 13: Reservoir required for a discharge at constant s.

These bounds hold *only* for a discharge at constant s. For a battery whose current falls to zero when the poles equalise (law L-bat of Section 10), s is nearly zero in the past and rises only recently; the data then allow s up to about 0.15 today and no bound of this type follows. What is immense, if anything, is the reservoir of condensate, not the antimatter: if the total charge is conserved, the twin contains as many antibaryons as we have baryons.

Other quantities the twin would influence: a_0 (through Γ), the ratio of the Ψ component to baryons (through co-creation), η (through the distribution of the charge). This requires the coupling law between the poles, the "internal resistance" of the battery, not yet derived.

11.3 The twin as the mirror half of the cycle (g)

The cycle solution is symmetric in time about the turnaround. If this mirror is also a matter–antimatter exchange (CPT), one recovers a twin made of antimatter, with a reversed arrow of time and zero total charge. Precedents: Gold (1962), an arrow of time that reverses at the turnaround (objection: entropy would decrease during the contraction); Boyle and Turok [8], a CPT mirror placed at the Big Bang rather than at the turnaround. Consistent, without an observable test today.

11.4 What would falsify or strengthen the idea

- False: $w < -1$ confirmed in the past (a battery does not recharge by itself); a_0 constant while the expansion evolves (the a_0–Γ link broken).
- Strengthened: a measured discharge ($w_0 > -1$) accompanied by an evolution of a_0 in the proportion of Eq. (31).

12 Primordial fluctuations without inflation

The compression of the cycle, dominated by matter, naturally produces a scale-invariant spectrum. With a single field, a theorem forbids an acceptable result; with two fields, the viable route is a recreation modulated by Ψ, which predicts a local non-Gaussianity $f_{\mathrm{NL}} \approx 5$.

Status (g). Wands' duality, the theorem of Li, Quintin and Cai, the relation $\zeta \approx -\frac{1}{6}\,\delta\Gamma/\Gamma$ and the result $f_{\mathrm{NL}} \approx 5$ are all established in GR. Section 13 states that at the bounce the emergent metric, hence GR, does not yet exist. This section therefore rests on the hypothesis that GR already applies to perturbations at the bounce. $f_{\mathrm{NL}} \approx 5$ also assumes a recreation rate linear in Ψ. Until these points are derived, P1 is a conditional prediction.

12.1 The matter bounce

A contraction dominated by matter automatically produces a scale-invariant spectrum: this is Wands' duality [64]. Omega does not need inflation; its cycle contains the mechanism. For $n_s = 0.965$, a slightly negative equation of state is needed during the contraction, $w \approx -0.003$ (not derived). Checked numerically: spectral slope 0.000 with constant sound speed.

12.2 The single-field no-go theorem

For a single-field matter bounce [37],

$$r = 24\, c_s, \qquad f_{\rm NL} \approx \frac{65}{8\, c_s^2} \quad (c_s \ll 1). \tag{34}$$

No value of c_s passes both. The theorem assumes a single field and GR at all energies, two

c_s during contraction	r (bound < 0.036)	$f_{\rm NL}$ (Planck: -0.9 ± 5.1)	verdict
1	24	$-35/8 \approx -4.4$ (canonical bounce)	excluded by r
10^{-3}	0.024	$\approx 8 \times 10^6$	excluded by $f_{\rm NL}$

assumptions Omega does not make; but this is a way out only if it is derived.

12.3 Recreation modulated by Ψ (g)

1. During compression, the clock and Ψ behave as dust; a light field in a matter-dominated contraction acquires scale-invariant fluctuations. Ψ therefore carries its own fluctuations.
2. At the bounce, the recreation rate of radiation, baryons and Ψ depends locally on Ψ: its fluctuations become density fluctuations, $\zeta \approx -\frac{1}{6}\,\delta\Gamma/\Gamma$.
3. Since everything is recreated together by co-creation, the perturbations are adiabatic, as Planck requires.

This is the modulated reheating of Dvali, Gruzinov and Zaldarriaga [17], transposed to recreation.

Why the conversion must be efficient. If $\zeta = \varepsilon\, \zeta_\Psi$ with an efficiency $\varepsilon \leq 1$ and an intrinsic non-Gaussianity f,

$$\zeta = \varepsilon\, \zeta_g + \frac{f}{\varepsilon}\, \zeta_g^2, \tag{35}$$

an inefficient conversion amplifies the non-Gaussianity instead of reducing it (as $f_{\rm NL} \approx 5/(4r)$ for a curvaton). A small sound speed of Ψ produces both amplified power (as c_s^{-3}, checked numerically) and $f_{\rm NL} \sim 1/c_s^2$. This route falls back under the theorem.

12.4 The viable route

Ψ has a sound speed close to c at the bounce. The suppression of r then comes from the conversion amplitude, not from c_s.

- The share of Ψ in ζ must exceed that of the clock by about 670 in power.
- For a recreation rate depending linearly on Ψ, the known result [70] is $\boldsymbol{f_{\rm NL} \approx 5}$**, of local shape**. Planck measures -0.9 ± 5.1 [52]: compatible at about 1.2σ.
- r is no longer predicted: it depends on the ratio of the two contributions.

12.5 Consequence: Ψ changes phase during the cycle

Ψ must have a sound speed close to c at the bounce and a negligible one at recombination and today (Section 9). An ordinary condensate law ($c_s \propto \sqrt{\rho}$) would give more than c at recombination: excluded. Ψ must therefore change phase between the bounce and recombination, according to cosmic time and density; nucleosynthesis requires the transition to occur before $1 + z_t \approx 7 \times 10^4$ if Ψ kept its energy at the transition.

12.6 Status

Element	status
Scale-invariant fluctuations	natural (matter-dominated contraction)
Adiabatic perturbations	from modulated co-creation
f_{NL}	≈ 5 (local shape), if $c_s(\Psi) \approx c$ at the bounce and GR applies to perturbations there
r	not predicted
n_s	requires $w \approx -0.003$ during contraction, not derived
$\eta \approx 6 \times 10^{-10}$	not derived

13 The CMB: echo of creation

The CMB is the echo of the constant creation of our dimension, prior to the appearance of GR. The measurements on its origin are compatible with this reading. The features of the sky (a fundamental, the quadrature between temperature and polarisation, all modes in phase, a perfectly Gaussian field) are the imprint of the resonance itself at our level; they are the targets the theory must reproduce.

13.1 A very old echo (o)

The CMB is a black body to 5×10^{-5} (FIRAS [23]). A mixture of black bodies whose temperatures are spread uniformly over a fractional width w has variance $w^2/12$ and produces a y-type distortion

$$y = \tfrac{1}{2} \left\langle \left(\frac{\Delta T}{T} \right)^2 \right\rangle = \frac{w^2}{24}. \tag{36}$$

The photons must therefore have been thermalised in a medium at a redshift of at least 2×10^6 (temperature $\gtrsim 0.5\,\mathrm{keV}$), a photon density about 10^{13} times that at recombination; this figure assumes the standard thermal history of the radiation era (M). This is exactly what is expected of an echo born in the dense phase: at the bounce the emergent metric, hence GR, does not yet exist, and the condensate is at its microscopic scale. The bounce density, at least $(2 \times 10^{16}\,\mathrm{GeV})^4$, is far above the threshold. The quasi-steady-state cosmology [27], which attributed the CMB to thermalised starlight, is refuted by this black body and by the acoustic peaks; Omega produces the CMB differently.

13.2 Present creation does not feed the CMB (o)

Measurements of the CMB temperature in the past (Sunyaev–Zel'dovich effect, molecular lines) give $T(z) = T_0(1+z)^{1-\beta}$ with $\beta = 0.010 \pm 0.013$: at most about 14 % of the CMB energy (re)created per e-fold of expansion today (2σ, the energy varying as T^4). Present creation

produces space, not the photons of the background. If the creation that drives the acceleration also produced the CMB, it would produce tens of thousands of times too much: the CMB channel accounts for at most a fraction of order 3×10^{-5} of the budget.

A resonance in motion, not a frozen image (g). The CMB is the resonance of perpetual creation, still active: the radiation bath keeps evolving (it cools with the expansion) and the medium in which it vibrates keeps being created. The pattern we receive carries the state of the vibration when the light left, as the sound of a bell struck without pause brings us the vibration of the moment it left the bell. The two measurements above ($T(z)$ and the exact correspondence between temperature and polarisation) bound this motion. Proposed direct test: distant clusters scatter the CMB they see from their own location, and the local quadrupole they receive can be read in their polarisation (polarised Sunyaev–Zel'dovich effect). A resonance that evolved otherwise than by simple cooling would show there.

13.3 Recombination is in the local-GR regime

At recombination the acceleration of the fluctuations is about $14\,a_0$ (Section 5): Omega reduces to GR there, and no structure exists yet. The height of the third peak (Section 9) and the origin of the fluctuations (Section 12) are questions distinct from the origin of the photons.

13.4 The peaks: targets of a resonance (o)

Positions of the extrema on the unbinned Planck 2018 spectra (reduced data (M), foregrounds removed by models, no standard cosmology assumed):

Spectrum	measured extrema
TT	peaks 222, 536, 814, 1123, 1415; troughs 415, 675, 1024, 1314
EE	peaks 396, 691, 1003; troughs 526, 836, 1161
TE	148, 308, 469, 596, 748, 915, 1071, 1223, 1376, 1535

- **A common fundamental.** A weighted linear fit of the extrema gives $\ell_A \approx 297.7$ (TT peaks), 299.6 (TT troughs), 296.8 (EE peaks), 311 (EE troughs), 303.6 (TE); an unweighted fit gives 297 to 318. The honest interval is $\ell_A \approx 297$–318, about 0.6° on the sky.
- **Quadrature.** The peaks of E fall on the troughs of T to within ± 0.07 period (396/415, 691/675, 1003/1024). The extrema of TE repeat at the half period (≈ 152). This is the signature of an oscillator and its derivative.
- **Offset equilibrium.** Successive TT spacings 193, 121, 139, 139, 210, 99, 191, 101; alternating heights. Phase of the TT peaks: $\ell = \ell_A(m - 0.24)$.
- **Phase coherence.** For peaks to exist, all modes of a given scale oscillate in phase, hence started at the same instant.
- Caveat: statistical errors (1 to 3 in ℓ) are smaller than the systematic shifts due to the damping envelope.

13.5 The fundamental is fixed by creation and by what the bounce created [d]

Computed with the background of Omega (vacuum created at the rate Γ, baryons, Ψ, photons and three neutrinos) and the oscillation of the plasma in the local-GR regime, the fundamental is

$$\ell_A = \pi \frac{D_M(z_*)}{r_s(z_*)} = 302.4 \qquad (z_* = 1\,092,\ r_s = 144.3\ \mathrm{Mpc},\ H_0 = 67.7\ \mathrm{km\,s^{-1}\,Mpc^{-1}}). \tag{37}$$

This is a consistency check, not a test: Γ is fitted on data that contain the CMB angle, ω_b and ω_Ψ come from the Planck fit, and closure imposes zero curvature; the result therefore recovers Planck's $\pi/\theta_* \approx 301.8$. Its sensitivities are $d\ln\ell_A/d\ln\omega_b = +0.05$, $d\ln\ell_A/d\ln\omega_\Psi = -0.15$, $d\ln\ell_A/d\ln\Gamma = -0.13$. Without Ψ, ℓ_A would be 460.

The resonance of the plasma is a resonance of the medium, but its fundamental is entirely fixed by Γ and by what the bounce created (baryons and charge of Ψ). Two consequences: $\ell_A \approx 300$ requires that Ψ gravitate in the expansion at recombination with $\omega_\Psi \approx 0.12$, independently of the peak heights; and a law for the unfolding of the resonance must not target ℓ_A directly, but the numbers fixed at the bounce (η, ω_Ψ/ω_b, the amplitude and tilt of the fluctuations, κ).

13.6 Gaussianity (o)

Reference: a Gaussian isotropic field with the measured spectrum, 500 simulations, $|b| > 30^\circ$, edges excluded. Skewness and kurtosis of the filtered maps in 8 bands from $\ell = 30$ to 1500: global $\chi^2 = 11.8$ for 16 statistics ($p = 0.76$). Excess kurtosis below 0.018–0.03 for $\ell > 400$ at 95 %.

Consequence for creation. If the echo comes from a discontinuous creation, each correlation area mixes at least 30 to 56 independent events (more than 7×10^7 events visible on the sky, with about 1.3×10^6 areas at $\ell \approx 1000$), or creation is continuous at this resolution. A superposition of a very large number of events naturally gives a Gaussian field.

13.7 Isotropy (o)

Template-free test. Spectrum $L = 1$ to 8 of the local-variance map (smoothed at 10°) in each band: no bin in excess below 1 %; a single one at $+2.9\sigma$ ($L = 1$, $\ell = 801$–1000), compatible with our motion. The variance of the fluctuations does not vary from one region of the sky to another.

Hemispherical asymmetry. Direction fixed in advance $(221^\circ, -20^\circ)$; asymmetry $A = (P_N - P_S)/(P_N + P_S)$. Final test with crossed half-missions (to cancel noise), $|b| > 30^\circ$, masks apodised at 2°, scales $\ell < 100$ removed before masking, 140 simulations: Beyond $\ell = 600$, only about

ℓ band	asymmetry (T)	significance
151–600	$+0.009 \pm 0.003$	2.8σ
601–1000	$+0.002 \pm 0.002$	0.8σ
1001–1500	$+0.002 \pm 0.002$	1.0σ

0.002 remains, the level expected from our own motion, consistent with Flender and Hotchkiss [24] ($A < 0.0045$ at 95 % for $\ell = 601$–2048). What remains is the known asymmetry, decreasing as roughly $\ell^{-1.3}$ and vanishing near $\ell \approx 600$; the four Planck component-separation methods agree. E-mode polarisation goes the same way ($+0.008 \pm 0.004$), not conclusive.

13.8 Reading of Omega (g)

The quadrature T/E, the offset equilibrium (an effect of the baryon loading) and phase coherence are also those of standard acoustics: phase coherence follows from adiabatic super-horizon

initial conditions, which the matter bounce of Section 12 already provides. The passage of the primary wave may be what triggers the resonance of the condensate at once, everywhere, but this is not the only possible reading: it is an interpretation (G), not demonstrated. The imprint of the primary wave is not a signal added to the sky: the sky itself, at the stage where we observe it, is that imprint. No geometry (planar or otherwise) applies at the level of the wave, since geometry exists only once GR has emerged. It remains to formulate how the state left by the wave in the condensate becomes a pattern on the sky when space emerges.

14 Black holes

In Omega, a black hole is the maximal concentration of forces at the Omega/GR boundary. Outside, Omega gives back exactly the Schwarzschild metric; inside, one reaches the condensate, without singularity. In its natural form, this sector gives no signal observable today: it is an open part of the theory, not a closed one.

14.1 Structure in the present framework (g)

- In the flow description of Section 4.2, the inflow speed is $v = c\sqrt{r_s/r}$, with $r_s = 2GM/c^2$. The horizon is the surface where the inflow of the condensate reaches c: an acoustic horizon. Since $g_{TT}g_{rr} = -c^2$ holds for any Φ, the exterior is exactly Schwarzschild, and the shadow seen by the Event Horizon Telescope is the same as in GR.
- Just inside, the condensate breaks at its microscopic scale ℓ, which the Crab bound constrains to $\lesssim 10^{-32}$ m. There is no singularity: the interior is the condensate itself.
- In the earlier formulation, black holes were breaking points of the condensate where boundary conditions quantise the modes, $E_k = \hbar c\, k/L_{\text{vib}}$ (discrete Hawking spectrum). The present framework keeps the idea of a discrete interior spectrum, but its observable consequences depend on the criterion below.

14.2 The decisive criterion: Landau's critical velocity [d]

A superfluid loses its superfluidity beyond a critical velocity $v_L \leq c$. Since the inflow reaches c at the horizon, the question is where the condensate breaks.

Waves of the condensate at very small scales	where the condensate breaks	consequence
Faster than c (natural case)	$v_L = c$, at the horizon	horizon preserved; no observable echoes. Omega differs from GR only inside (no singularity) and in the Hawking radiation, unobservable for astrophysical holes
Slower than c, $v_L = c(1-\epsilon)$	outside the horizon	object with a surface. If $\epsilon \sim 1$: shadow and ringdown modified, excluded. If ϵ tiny (10^{-15}–10^{-20}, fine-tuned): gravitational-wave echoes at 0.04–0.06 s for GW150914 and 0.09–0.13 s for GW190521

In the second case, the reflecting cavity between the photon sphere and the breaking surface would have discrete modes and would return echoes with a delay of about $(8GM/c^3)\ln(2r_s/\ell)$. This case runs into the absence of thermal surface emission from Sgr A* [9] and into the instability of rotating objects without a horizon. Omega therefore predicts, in its natural form,

no echoes. Searches by the LIGO–Virgo–KAGRA collaboration have found no significant echo; earlier claims of marginal echoes near 0.3 s for GW150914 [1] are disputed.

14.3 Cosmological coupling

The condensate flows into black holes. If it has the density of the vacuum energy, a black hole of $10^9 M_\odot$ absorbs only about $4 \times 10^{-5} M_\odot$ in a time $1/H$. Omega therefore does not predict the growth of black-hole masses with the expansion claimed by Farrah et al. [22], a claim that is itself disputed.

14.4 The boundary far from the black hole (g)

If a black hole is a concentration at the boundary, the boundary itself, where $g \approx a_0$, and not the horizon, could leave a signature. It lies at

$$r_M = \sqrt{\frac{GM}{a_0}} : \qquad \text{about 70 pc for Sgr A* } (4 \times 10^6 M_\odot), \text{ about 3 kpc for M87* } (6.5 \times 10^9 M_\odot), \tag{38}$$

counting only the mass of the black hole. This follows the same logic as the cluster excess that dies out where $g \approx a_0$ (P5 bis). It is untested, and no quantitative prediction is yet attached to it.

14.5 Status

- Derived: exterior exactly Schwarzschild; horizon as the surface where the inflow reaches c; no singularity; no echoes in the natural case.
- Open: the thermodynamics of black holes (entropy and temperature of a condensate horizon); rotating black holes (the river model of Kerr requires a torsion that the present description does not explain); the signature of the boundary at r_M.
- The massive modes of the condensate (above 5 meV) would act near 10^{12} Hz, out of reach.

15 The quantum level and the unfolding of the resonance

The quantum world is the first level of the unfolding of the resonance left by the primary wave, before GR. It is the least developed axis of Omega and probably the one where a prediction specific to the theory will be found.

15.1 The scheme (g)

- The dimensional pulsation crosses the scalar field and creates a resonance in it.
- This resonance organises itself into a structure at the sub-quantum level.
- It unfolds into increasingly organised levels: quantum, then structures, up to our universe, where GR appears locally.
- Earlier formulations described a "supra-cosmic condensate activated by a pulse", vibrational modes φ_k and a spectral flow admitting travelling fronts (Fisher–KPP type). Their vibration equation, with a threshold term and a fractional memory, is not yet connected to the action of the present architecture.

15.2 A resonance can produce quantisation

The walking droplets of Couder and Fort [14] show experimentally that a wave of a medium guiding a particle produces quantised orbits, interference and tunnelling-like effects. Omega would be the fundamental version, with the condensate as the medium and the primary wave as the source of the resonance. In a superfluid, circulation is already quantised (h/m); the local limit of a scalar condensate needed exactly this quantisation to reproduce frame dragging (Section 4).

15.3 The rhythm of Ψ and the bridge between scales

- The proper rhythm of Ψ is fixed by observation: $\hbar Q_0 \approx 5\,\mathrm{GeV}$, about $10^{24}\,\mathrm{Hz}$ (Section 9). The cycle is another, slow rhythm of the whole condensate.
- A cascade in frequency of Fisher–KPP type would transport energy from one frequency to another with a front of constant speed $v_{\min} = 2\sqrt{D\lambda}$. From the rhythm of Ψ ($10^{24}\,\mathrm{Hz}$) to the frequencies of structures (about $10^{-17}\,\mathrm{s}^{-1}$) there are about $\ln(10^{41}) \approx 94$ e-folds, the same order as the 67 e-folds between bounce and turnaround. Plausible, not derived.
- A direct resonance between $10^{24}\,\mathrm{Hz}$ and dynamical rates ($\sim 10^{-16}\,\mathrm{s}^{-1}$) is impossible; a beat or a cascade is needed.

Local deposition criteria (o). If the resonance deposited the energy of the rhythm of Ψ according to a single local quantity, that quantity should separate the systems that require a deposit from those that do not: For each criterion, a "no" system falls between two "yes". The

System	V/R	ratio to expansion	density ($\mathrm{kg/m^3}$)	temperature	deposit required
CMB (peaks)	$4 \times 10^{-13}\,\mathrm{s}^{-1}$	8	4×10^{-18}	3 000 K	yes
Lyman-α, $z \approx 3$	1×10^{-17}	1	2×10^{-24}	2×10^4 K	yes
Clusters (R_{2500})	7×10^{-17}	30	2×10^{-23}	6×10^7 K	yes
Groups	2×10^{-17}	7	1×10^{-24}	1×10^7 K	no
Galaxy outskirts	3×10^{-16}	150	3×10^{-22}	5×10^3 K	no

resonance cannot rest on a single local quantity: it depends on the history of the system, in line with the principle that each body has its own rules. A non-uniform primary wave produces exactly such a history dependence.

15.4 A quantifiable bridge between quantum and cosmic

If the same unfolding fixes the quantum ($\hbar$, the masses) and the cosmic (Γ, a_0), these quantities must be related.

- Known coincidence (Weinberg [65]): $(\hbar^2 H/Gc)^{1/3} \approx 100\,\mathrm{MeV}$, the hadronic scale. It reads as the equality between the gravitational energy of a mass at its own Compton length and the quantum of energy $\hbar H$:

$$\frac{Gm^2}{\hbar/(mc)} = \hbar H \quad \Rightarrow \quad m^3 = \frac{\hbar^2 H}{Gc}. \tag{39}$$

- Declared contamination: with the values of Omega this gives about 56 MeV with $H_\infty = \Gamma/3$ and about 81 MeV with Γ, computed before any law. A law that only recovered this scale would not count; it would have to fix the exact factor without adjustment.

- For standard physics this is an unexplained coincidence; for Omega it would be a required consequence. The co-creation charge $\hbar Q_0 \approx 5\,\mathrm{GeV}$ already ties $\hbar$ to creation.

15.5 The central prediction to be built

If a single resonance generates all levels, its characteristic scales are tied by the law of the unfolding. The method is fixed in advance: the law is written before any computation, with no parameter added, and it is retained only if it recovers at least two independent quantities to better than 10 %; a failure is recorded. The targets are the numbers fixed at the bounce: $\eta \approx 6.1 \times 10^{-10}$; $\omega_\Psi/\omega_b = 5.364 \pm 0.065$ (that is $\hbar Q_0 = 5.03\,\mathrm{GeV}$); the amplitude $A_s \approx 2.1 \times 10^{-9}$ and tilt $n_s \approx 0.965$ of the fluctuations; and κ.

A cascade has at least two internal parameters; tuning them on η and on the ratio 5.36 would predict nothing. Its parameters must therefore be fixed by known quantities: its rate by Γ, its starting point by the microscopic scale of the condensate. The only dimensionless number available is then the hierarchy

$$N = \ln \frac{E_*}{\hbar\Gamma} = 139.4 \;\; (E_* = E_{\mathrm{Planck}}), \qquad 133.0 \;\; (E_* = 2 \times 10^{16}\,\mathrm{GeV},\ \mathrm{Crab}), \tag{40}$$

and a law without tuning must take the form

$$\eta = e^{-\alpha N}, \qquad \frac{\hbar Q_0}{E_*} = e^{-\beta N}, \tag{41}$$

where α and β must follow from the structure of the cascade (number of levels, dimension, order of the process). Targets: $\alpha = 0.152$, $\beta = 0.304$ with the Planck energy; $\alpha = 0.160$, $\beta = 0.270$ with the Crab scale. The near equality $\hbar Q_0 \approx \eta^2 E_{\mathrm{Pl}}$ (4.54 against 5.03 GeV) was noticed after the computation; it depends on the choice of reference scale and misses by 10 % quantities known to 1 %. It is recorded as a coincidence, not a result.

A predictive cascade law therefore first requires a microscopic model of the condensate that fixes, by physics and before any computation, the starting scale and the structure of the levels. This is the present lock of the central prediction.

16 Discussion

16.1 What is derived and what is postulated

16.2 Relation to other frameworks

- **ΛCDM.** The expansion history of Omega is, by construction, that of flat ΛCDM, and in the linear cosmological regime the Ψ component gravitates as cold dark matter. The difference is in the interpretation (creation of the medium instead of a dark-energy fluid; a field rhythm fixed by co-creation instead of particles), in galaxies (no halo, low-acceleration regime), and in the link between a_0 and the vacuum.
- **MOND** [44]. Omega recovers its phenomenology in galaxies and adds a physical origin for a_0 (the creation rate), a fixed shape constraint ($p \geq 1$), a memory time, and a time evolution of a_0 tied to the vacuum.
- **AeST** [58]. The field Ψ of Omega has exactly the structure of the scalar sector of AeST; we credit it fully. What is specific to Omega: the emergence of GR from a tensor condensate (cardinal gravity) rather than a postulated metric; creation by the rolling vacuum; the link a_0–Γ and its consequence P3 bis; the fixed relation of the pressureless component to the baryon charge; the cycle and the twin as a battery.

Result	status	section
Schwarzschild from the superposition of flow energies; Mercury; $\gamma = 1$; lensing = dynamics	[D] under (G) hypotheses	4.2
Tensor order required by gravitational-wave polarisations	[D] + (O)	3
Exponent 3/2, $V^4 = GMa_0$, $MGa_0 = \frac{81}{4}\sigma^4$	[D]	5, 6
Fast transition ($p \geq 1$) and memory τ	imposed by Saturn and Cassini [N]	5
a_0 does not follow $H(z)$	(M) (RC100)	6
de Sitter expansion at zero energy, $H_\infty = \Gamma/3$	[D] from $V(A)$ (postulated)	8.3
Braking by matter reproducing ΛCDM distances	(G), energy balance; not yet from the full tensor dynamics	8
One clock cannot give dust and the low-acceleration response	[D]	9.2
Pressureless component from the rhythm of Ψ	[D] from shift symmetry	9
$\hbar Q_0 = 5.03\,\mathrm{GeV}$	relation from co-creation (G); amount taken from data	9
Linearity identity, stiffening of Ψ	[D]	9.5
Cycle (turnaround, bounce, periodic episodes)	[D] [N] from $V(A)$ (postulated)	10
a_0 follows the vacuum; relation P3 bis	structural argument (G)	8.6, 10.5
$f_{\mathrm{NL}} \approx 5$	conditional (GR at the bounce, linear recreation)	12
$\ell_A = 302.4$	consistency check, not a test	13

Table 14: Status of the main results.

- **Khronon and Hořava gravity** [6, 48]. Omega shares the preferred time and the geometric dust, and shows that a single clock cannot carry both the dust and the low-acceleration response.
- **Emergent gravity** [61]. Verlinde obtains $a_0 \approx cH_0/6$; in readings where a_0 follows $H(z)$, it is excluded by RC100. In Omega a_0 follows the vacuum alone.
- **Bouncing cosmologies** [64, 3, 8]. The matter bounce, the $-1/a^8$ correction and the CPT mirror have precedents; in Omega they follow from the dilation potential of the condensate.

16.3 Open problems

1. κ: not derived; every natural mechanism gives a factor of order one instead of 0.2 (Section 8).
2. $\hbar Q_0$, η, **the barrier against the conversion of Ψ into nucleons**: not derived (Section 9).
3. **Capture of Ψ in clusters but not in galaxies**: mechanism not found (Sections 7, 9).

4. **Baryonic Tully–Fisher slope**: 4 predicted, 3.62 ± 0.09 in our fit, about 3.85 in the literature.
5. **Ultra-faint dwarf galaxies**: fail for every model in which the gravity excess is tied to the baryons.
6. **Braking by matter** from the full kinetic term of the tensor, and the moment of the turnaround (Section 10).
7. **Survival of the cardinal-gravity bootstrap** around the rolling vacuum.
8. **Memory**: relate τ to Γ ($\tau\Gamma \approx 3$–9×10^{-5}).
9. **Compression phase**: n_s ($w \approx -0.003$), the change of phase of Ψ, GR at the bounce.
10. **CMB**: how the state left by the wave becomes a pattern on the sky when space emerges.
11. **Black holes**: thermodynamics, rotation, the boundary at r_M.
12. **Universality of the emergent metric**: posited, not derived.
13. **DESI**: if a crossing of $w = -1$ in the past is confirmed, the battery and the cycle are in difficulty (P2).

16.4 Earlier formulations

Omega went through several published versions (OSF project osf.io/sjndw) and many intermediate branches. Their predictions specific to the theory were confronted with data and are not retained: a non-expanding universe with a frequency-dependent redshift (excluded by the time dilation of supernova light curves, $(1+z)^b$ with $b = 1.003 \pm 0.005 \pm 0.010$ [67]); a spectral slope of gamma-ray burst light curves; an outer slope $v \propto r^{-1/4}$ of rotation curves (excluded by SPARC); a refractive index for gravitational waves (excluded by GW170817); a scalar–tensor gravity in which lensing is not enhanced (excluded by clusters). The present architecture was built under these constraints. All negative results are documented in the project archive.

17 Predictions

The predictions below were first recorded in a dated register on 21 and 22 October 2026. Rules: a prediction is never modified after deposit; a revision is added as a new dated entry. Each entry states whether it is **specific to Omega** or **shared** with other models: a shared prediction can falsify Omega but cannot validate it on its own.

17.1 Table of predictions

The first group contains the predictions specific to Omega; the second, the predictions shared with other models.

No.	Prediction	Decisive data	Falsified if	Status
Predictions specific to Omega				
P3 bis	a_0 follows the dilation rate of the vacuum. Its evolution is fixed by the present equation of state of the vacuum, without free parameter: $a_0(z \gtrsim 1)/a_0(0) = \sqrt{1+\frac{3}{4}(1+w_0)}$ (turnaround law) or $1+\frac{1}{2}(1+w_0)$ (battery law). At most 5 % with present bounds; 9 to 12.5 % if $w_0 \approx -0.75$.	Final DESI and Euclid for w_0; galaxy kinematics at $z \approx 1$–2 (JWST, ELT, successors of RC100) for a_0. Present precision on $a_0(z=2)/a_0(0)$: $[0.76; 1.18]$, not sufficient.	$w_0 > -1$ confirmed without crossing -1, and a_0 unchanged at $z \approx 1$–2 to better than half the predicted excess; or a_0 evolving while $w_0 = -1$.	(G); structural argument (Section 8.6).
P3	a_0 is locked to the creation rate Γ ($a_0 = \kappa c \Gamma$), not to the expansion rate $H(z)$.	High-redshift rotation curves.	a_0 grows with z as $H(z)$ (1.7 to 3 times larger at $z = 1$–2).	(G); consistent with RC100 ($a_0 \propto H(z)$ rejected at $\Delta\chi^2 = +83$). The constancy of a_0 is shared with MOND; the lock to Γ is specific.
P5 bis	In a galaxy cluster, the mass excess is confined inside the boundary and dies out where the observed acceleration falls to a_0. The cut-off radius follows, cluster by cluster, the radius where $g = a_0$ ($\propto T$ for an isothermal cluster), not a fixed fraction of an overdensity radius ($\propto T^{1/2}/E(z)$).	Lensing and X-ray profiles of clusters spanning at least a factor 10 in mass (eROSITA, CHEX-MATE, Euclid).	The cut-off follows a fixed fraction of R_{500}, independently of mass.	(G); first test on 16 CLASH clusters not informative.
Shared predictions				

No.	Prediction	Decisive data	Falsified if	Status
P2	The equation of state of the vacuum never goes below -1: $w \geq -1$ at all epochs, with $1+w = \frac{4}{3}x^4/(1-x^4)$ or $2x^3/(1-x^3)$. Bounds (2σ): $a_{\max} \geq 1.81$ or $a_{\mathrm{eq}} \geq 2.74$.	DESI final analyses, Euclid; non-parametric reconstruction of $w(z)$ by redshift bins.	$w < -1$ at more than 3σ in at least one bin.	(G); shared with non-phantom dark-energy models. Under pressure: the w_0–w_a fits of DESI DR2 prefer an evolution at 2.8 to 4.2σ.
P1	Primordial non-Gaussianity of local shape from the recreation modulated by Ψ at the bounce: $f_{\mathrm{NL}} \approx 5$.	SPHEREx, then galaxy surveys; target precision ± 0.5 to ± 1.	f_{NL} compatible with 0 and values near 5 excluded at more than 3σ.	(G); conditional (GR applies to perturbations at the bounce; recreation linear in Ψ); shared with modulated reheating.
P4	Gravitational waves carried by a tensor condensate: speed exactly c; two tensor polarisations only; additional modes massive (above about 5 meV if they couple with gravitational strength, otherwise above 10^{-12} eV).	LIGO–Virgo–KAGRA, then LISA.	A scalar or vector polarisation is detected, or a deviation from c.	(G); shared with GR (consistency).
P6	The component carried by the rhythm of Ψ is not made of particles and acts only through gravity: no direct detection, no annihilation or production signal, whatever the sensitivity.	Underground detectors (XENONnT, LZ and successors), gamma-ray telescopes, LHC.	Confirmed detection of a dark-matter particle.	(G); shared with every theory without dark-matter particles; consistent with all searches to date.
C1	Black holes: exterior exactly Schwarzschild (shadow as in GR); no gravitational-wave echoes in the natural case; no cosmological growth of black-hole masses with the expansion.	EHT, LIGO–Virgo–KAGRA ringdowns, black-hole demographics.	Confirmed echoes, or a cosmological mass growth $M \propto a^k$ with $k > 0$.	[D] under (G); shared with GR.

No.	Prediction	Decisive data	Falsified if	Status
C2	No variation of the gravitational constant: $\dot{G} = 0$.	Lunar laser ranging, pulsars.	$\dot{G} \neq 0$ detected.	(G); shared with GR.
C3	Baryonic Tully–Fisher slope exactly 4 ($V^4 = GMa_0$); lensing mass equal to dynamical mass at all accelerations.	SPARC-like samples, weak lensing (KiDS, Euclid).	A slope significantly different from 4 after systematics; lensing and dynamics diverging around galaxies.	[D]; shared with MOND. In tension with our SPARC fit (3.62 ± 0.09).
C4	The CMB temperature scales as $T_0(1+z)$: present creation does not feed the CMB ($\beta = 0$).	SZ and molecular-line measurements of $T(z)$.	$\beta \neq 0$ detected.	(O) consistent ($\beta = 0.010 \pm 0.013$); shared.
C5	The range μ^{-1} of the mass term of Ψ is shorter in clusters than around galaxies and does not increase with redshift.	Galaxy–galaxy and cluster lensing in redshift bins (Euclid, LSST), once the deviation is detected.	μ^{-1} larger at high redshift, or equal in clusters and galaxies.	[D] qualitative; shared with AeST (stiffening forms); amplitude not fixed.

17.2 Axes without a quantitative prediction yet

- **Black holes:** a possible signature of the boundary far from the hole, at $r_M = \sqrt{GM/a_0}$ (about 70 pc for Sgr A*, 3 kpc for M87*); thermodynamics; rotation.
- **Quantum level:** the law of unfolding of the resonance, which should tie η, $\hbar Q_0$, the amplitude of the fluctuations and κ without adjustable parameter (Section 15). This is the candidate for the central prediction of Omega.
- **CMB:** how the state left by the primary wave becomes the pattern on the sky; the moving resonance tested through the polarised Sunyaev–Zel'dovich effect of distant clusters.
- κ**:** a derivation would turn the coincidence $a_0 \approx 0.2\, cH_\infty$ into a prediction, the analogue of the perihelion of Mercury for GR.

17.3 Conclusion

Omega is a falsifiable framework in which GR is a local limit of a continuously created condensate. It reproduces the local tests of gravity, the dynamics of galaxies and groups, the expansion history, the linear growth probed by the Lyman-α forest and the structure of the CMB, with declared open problems (κ, the amount of the Ψ component, clusters, the Tully–Fisher slope, ultra-faint dwarfs). Its first prediction specific to the theory, P3 bis, ties the evolution of the galactic acceleration scale to the equation of state of the vacuum without any free parameter. It will be tested when the vacuum is shown to depart from $w = -1$ and when a_0 is measured at $z \approx 1$–2 to a few per cent.

Quantity	value	origin
Creation rate Γ	$5.46 \times 10^{-18}\,\mathrm{s}^{-1}$ ($1/\Gamma \approx 5.8\,\mathrm{Gyr}$)	fit of the expansion (M)
Asymptotic expansion H_∞	$\Gamma/3$ ($1/H_\infty \approx 17.4\,\mathrm{Gyr}$)	rolling vacuum (G)
Dilation mass m	$2\Gamma/3$; $\hbar m \approx 2.4 \times 10^{-33}\,\mathrm{eV}$	rolling vacuum
a_0	$1.16 \times 10^{-10}\,\mathrm{m\,s}^{-2}$ (McGaugh form); 1.37–1.56 with series forms	SPARC (M)
$\kappa = a_0/(c\Gamma)$	0.055–0.08 (McGaugh, with systematics); 0.084–0.095 (series forms allowed by Saturn)	fitted, not derived
$a_0/(cH_\infty)$	0.16–0.24	depending on H and form
Co-creation energy $\hbar Q_0$	$5.03 \pm 0.06\,\mathrm{GeV}$	ω_Ψ/ω_b (M)
Memory time τ	0.15–0.5 Myr	Cassini, wide binaries
Transition exponent	$p \geq 1$	Saturn
Mass-term range μ^{-1}	$\gtrsim 1\,\mathrm{Mpc}$ today around galaxies	galaxies, clusters
Additional tensor modes	$\gtrsim 5\,\mathrm{meV}$ (gravitational coupling)	torsion balances
Preferred-frame residual	$\alpha \lesssim 10^{-7}$	solar system, pulsars
Turnaround	$a_{\max} \geq 1.81$ (L-ret); $a_{\mathrm{eq}} \geq 2.74$ (L-bat)	expansion data, 2σ
Drift of Γ (power law)	$\lvert d\ln\Gamma/dt\rvert \lesssim 0.03\,H_\infty$	expansion data, 2σ
Healing length of the condensate	$\lesssim 10^{-32}\,\mathrm{m}$	Crab nebula
CMB fundamental ℓ_A	297–318 measured; 302.4 computed (consistency)	Planck 2018

Table 16: Reference values.

A Reference values and notation

Notation. u^μ: timelike eigen-direction of the condensate (universal clock). $A(t)$: spatial eigenvalue of the tensor condensate, $a^2 \propto A$. $V(A)$: dilation potential, Eq. (2). $Q = u^\mu \partial_\mu \Psi$: rhythm of Ψ; $Y = |\nabla_\perp \Psi|^2$: its gradient; $\Pi = K'(Q)$: its charge; $\chi = K''$: its susceptibility. μ: mass term of Ψ in the static limit. $x = g/a_0$ in Section 5; $x = a/a_{\max}$ in Section 10. $r_M = \sqrt{GM/a_0}$: radius of the Omega/GR boundary of a point mass.

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